Graphical Abstract

Vector Nodal Meshless Method (VNMM) for Electromagnetic Problems: Support Stabilization, Complete Bases, and Conforming Hybrid EFEM Coupling

Renato C. Mesquita, Elson J. da Silva, Bárbara M. F. Gonçalves, Luilly A. G. Ortiz

Highlights

Vector Nodal Meshless Method (VNMM) for Electromagnetic Problems: Support Stabilization, Complete Bases, and Conforming Hybrid EFEM Coupling

Renato C. Mesquita, Elson J. da Silva, Bárbara M. F. Gonçalves, Luilly A. G. Ortiz

- Complete linear basis $\mathcal{P}^1$ removes derivative modal aliasing in 2D VNMM.
- Breaks the $\mathcal{O}(1)$ curl stagnation, restoring $\mathcal{O}(h)$ asymptotic convergence.
- Scale-dependent determinant thresholding stabilizes stencils under 25% jitter.
- Div-curl Ritz-Galerkin penalty shifts spurious gradient modes outside spectrum.
- Master-slave hybrid coupling eliminates saddle-point indefiniteness without multipliers.

Vector Nodal Meshless Method (VNMM) for Electromagnetic Problems: Support Stabilization, Complete Bases, and Conforming Hybrid EFEM Coupling

Renato C. Mesquita[a,*], Elson J. da Silva[a], Bárbara M. F. Gonçalves[a], Luilly A. G. Ortiz[a]

[a] *Department of Electrical Engineering, Universidade Federal de Minas Gerais, Av. Antônio Carlos, 6627, Belo Horizonte, 31270-901, MG, Brazil*

Abstract

The Vector Nodal Meshless Method (VNMM) provides an attractive mesh-free alternative to the Edge Finite Element Method (EFEM) by defining vector approximations within $H(\mathbf{curl})$ Sobolev spaces through point-collocated nodes assigned unit vector directors. However, conventional formulations based on Nédélec's incomplete solenoidal polynomial space ($\mathcal{L}^1$) suffer from a severe mathematical limitation when operating on arbitrary point clouds: numerical approximation of the curl operator ($\nabla \times \vec{E}$) fails to converge, stagnating at an irreducible $\mathcal{O}(1)$ relative error plateau.

This paper presents a stabilized 2D VNMM framework that addresses this limitation and provides a general vector approximation formulation for mesh-independent point distributions. First, we introduce the complete linear polynomial vector space $\mathcal{P}^1 = \mathcal{P}_1 \times \mathcal{P}_1$ (6 support nodes per evaluation point), which resolves all four spatial components of the local Taylor Jacobian tensor. This removes the derivative modal aliasing mechanism, breaking the $\mathcal{O}(1)$ stagnation plateau, restoring first-order asymptotic convergence $\mathcal{O}(h^1)$ for the curl operator, and yielding an observed approximately quadratic convergence rate $\mathcal{O}(h^2)$ for electric field interpolation in the investigated benchmark.

Second, to ensure algebraic stability across arbitrary node clouds, we de-

*Corresponding author

Email address: renato@ufmg.br (Renato C. Mesquita)

velop a 4-phase automated support selection algorithm combining k-d tree spatial queries with scale-dependent local placement matrix $\mathbf{A}$ determinant thresholding. Enforcing an absolute tolerance floor provides a unified algebraic criterion for rejecting ill-conditioned local placement configurations, maintaining bounded condition numbers ($\kappa(\mathbf{A}) \leq 73$) and restoring monotonic error reduction under a representative 25% stochastic spatial jitter with random unit director orientations.

Third, to mitigate non-physical spurious modes caused by discrete gradient nullspace leakage ($\nabla \times (\nabla\phi) \neq \mathbf{0}$ at quadrature points), we implement a Ritz-Galerkin weak formulation with div-curl penalty regularization and point-based Element-Free Galerkin (EFG) Gauss integration. This shifts non-physical gradient states outside the operational frequency window, yielding spurious-free modal pairing for the investigated transverse electric cavity resonances (TE_z) with a 1.00% mean wavenumber error across the tested band.

Finally, we establish a conforming master-slave hybrid EFEM-VNMM domain decomposition solver based on the exact dimensional scale transformation. This direct reduction enforces interface tangential continuity without auxiliary Lagrange multipliers, avoiding zero-mass saddle-point indefiniteness, preserving symmetric positive-definite (SPD) system properties under regularized boundary conditions, and preventing spurious infinite eigenvalues ($\lambda = \infty$) in the discrete spectrum.

Crucially, the proposed formulation liberates the VNMM from the topological constraints of prior literature, eliminating the requirement for auxiliary planar subdivisions or Delaunay triangulations and enabling robust electromagnetic analysis on unstructured vector node distributions.

Keywords: Meshless methods, Vector Nodal Meshless Method (VNMM), Curl-conforming, Conforming hybrid coupling, Spurious modes

---

1. Introduction

This section provides the contextual framework, fundamental motivations, and key scientific challenges that drive the development of advanced meshless methods for computational electromagnetics. First, the historical evolution and practical advantages of meshless paradigms over traditional grid-based methods are outlined. Subsequently, the mathematical bottlenecks of earlier vector meshless formulations—specifically derivative stagnation and support

instability—are detailed, highlighting the necessity for the stabilized Vector Nodal Meshless Method (VNMM 2D) and the conforming hybrid coupling strategy proposed in this work.

1.1. Context and Motivation

The mathematical modeling of electromagnetic phenomena governed by Maxwell's partial differential equations plays a pivotal role in modern science and engineering applications [1, 2]. Because exact analytical solutions are strictly restricted to canonical geometries and homogeneous media, numerical discretization methods have become indispensable tools for simulating complex electromagnetic devices, waveguiding structures, and resonant cavities [1, 3].

Historically, spatial discretization in computational electromagnetics (CEM) originated with finite difference techniques, notably Yee's staggered Finite-Difference Time-Domain (FDTD) grid [4], and the conventional node-based scalar Finite Element Method (FEM) [1]. In node-based FEM formulations, vector field variables (such as the electric field $\vec{E}$ or magnetic field $\vec{H}$) are discretized by assigning independent scalar degrees of freedom (DoFs) to each Cartesian component (E_x, E_y, E_z) at shared mesh vertices [1, 5]. While highly successful for scalar Helmholtz and Poisson problems, applying node-based scalar approximations directly to vector Maxwell equations leads to fundamental mathematical pathologies [5, 6]:

1. **Violation of Tangential Interface Continuity:** Physical vector fields across dielectric or magnetic material boundaries require continuous tangential components ($\mathbf{n} \times [\vec{E}] = \mathbf{0}$) while allowing normal components to jump discontinuously [1, 3]. Node-based scalar shape functions enforce C^0 continuity across all vector components simultaneously, artificially locking normal jumps or under-specifying tangential continuity [5].
2. **Field Singularities at Sharp Edges:** Node-based approximations fail to correctly capture singular field behavior near sharp conducting re-entrant corners and edges [3, 6].
3. **Spectral Contamination by Spurious Modes:** Continuous scalar approximations do not satisfy the exact discrete de Rham complex property ($\mathrm{Im}(\nabla) \subset \ker(\nabla\times)$) [3, 5]. Consequently, the infinite-dimensional nullspace of irrotational gradient fields ($\vec{E} = \nabla\phi$) leaks into the physical discrete spectrum, generating non-physical, highly oscillatory *spurious modes* interspersed among true physical resonances [5, 6].

To overcome these severe flaws, Nédélec [7] introduced mixed vector finite elements—commonly designated as Whitney 1-forms or Edge Finite Element Methods (EFEM) [3, 8]. As extensively detailed by Bossavit [8], by associating degrees of freedom with line integrals of field circulation $e_k = \int_{E_k} \vec{E} \cdot d\boldsymbol{\ell}$ along element edges rather than scalar nodal points, EFEM constructs vector shape functions natively conforming in the Sobolev space $H(\mathbf{curl}; \Omega)$. EFEM naturally satisfies tangential interface continuity, models edge field singularities, and preserves the discrete de Rham sequence, completely eliminating spurious modes from electromagnetic spectrum calculations [3, 5, 8].

Despite the undeniable success of EFEM, all mesh-based formulations remain heavily burdened by the pre-requisite of generating high-quality 3D conformal meshes [9, 10]. For problems involving complex CAD geometries, moving interfaces (such as rotating electrical machinery), severe spatial deformations, or adaptive mesh refinement, the computational overhead of mesh generation, remeshing, and element distortion repair represents a major bottleneck [9, 10, 11].

To liberate numerical solvers from topological mesh constraints, *Meshless (or Meshfree) Methods* emerged as a powerful computational paradigm in solid mechanics, fluid dynamics, and thermodynamics [10, 12, 13]. By constructing field approximations directly over arbitrary clouds of points without pre-arranged element connectivity, meshless methods eliminate mesh generation costs and facilitate seamless adaptive node insertion [10, 14]. Prominent weak-form meshless paradigms include the Element-Free Galerkin (EFG) method [14], based on Moving Least Squares (MLS) approximations over background quadrature cells; the Meshless Local Petrov-Galerkin (MLPG) method [15, 13], utilizing local quadrature subdomains to achieve a "truly meshless" formulation; the Point Interpolation Method (PIM) and Radial Point Interpolation Method (RPIM) [16, 12]; Smoothed Particle Hydrodynamics (SPH) [17]; and the Reproducing Kernel Particle Method (RKPM) [10].

Meshless formulations have demonstrated significant advantages over traditional grid-based methods in low-frequency electromagnetics, particularly for modeling deforming domains, moving boundaries, and transient phenomena in electrical machines without requiring complex remeshing procedures [18, 19, 20]. Early attempts to adapt meshless techniques to computational electromagnetics—such as MLS-RKPM formulations [18], EFG schemes [21, 19], and MLPG vector formulations [13, 22, 23]—primarily

constructed scalar meshless shape functions for individual vector components (E_x, E_y, E_z). However, treating vector components independently inherited all the historical pathologies of scalar node-based FEM: appearance of non-physical spurious modes, difficult imposition of tangential interface conditions across material discontinuities, and inflated system matrix dimensions requiring auxiliary divergence-free constraints or Lagrange multipliers [24, 25, 26].

To establish a truly $H(\mathbf{curl})$-conforming meshless vector framework, Lima and Mesquita [27] developed the *Edge Meshless Method* (EMM). The EMM distributes directed spatial line segments (edges) throughout the domain and constructs vector shape functions based on $H(\mathbf{curl})$ space properties [27, 25].

Serving as a crucial bridge toward higher-order vector meshless representations, Silva, Mesquita, and Lima [28] introduced $H(\mathbf{curl})$-conforming vector shape functions for the EMM based on Nédélec's incomplete solenoidal $\mathcal{L}^1$ polynomial space [7]. By refining local circulation projections along support edges, their formulation ensured exact tangential field continuity across material interfaces, divergence-free behavior, and complete suppression of spurious modes. Building upon these vectorial shape function concepts, Ortiz et al. [29, 30, 26] established higher-order expansions and investigated the asymptotic limit as physical edge lengths shrink to zero ($L_k \to 0$), giving rise to the Vector Nodal Meshless Method (VNMM 2D).

In VNMM, the 1D continuous edge degenerates into a point-collocated vector node $(P_i, \vec{t_i})$, where each node P_i is assigned a unit vector director $\vec{t_i}$ [30, 26]. On physical boundaries and material interfaces, $\vec{t_i}$ is aligned strictly tangent to the boundary, enabling direct, unconstrained imposition of essential PEC conditions and tangential field continuity [30, 31]. By associating a single scalar directional projection DoF ($c_i = \vec{E}(P_i) \cdot \vec{t_i}$) with each node, VNMM combines the lightweight single-DoF-per-node efficiency of scalar meshless methods with the $H(\mathbf{curl})$ conforming properties of Whitney edge elements [30, 32, 31]. Subsequent developments extended the VNMM to three-dimensional vector problems [32, 31] and periodic electromagnetic structures [33].

To synthesize these fundamental distinctions and contextualize the VNMM within the broader landscape of computational electromagnetics and meshless research [10, 14, 13, 11], Table 1 summarizes the mathematical and algorithmic characteristics of prominent grid-based and meshless schemes, contrasting their variable representations, interface continuities, spurious-mode

control mechanisms, and curl convergence behaviors.

1.2. The Curl Stagnation Bottleneck in Arbitrary Supports

Despite the mathematical elegance of the original Vector Nodal Meshless Method (VNMM 2D) [30, 26], early formulations were fundamentally tied to structured support selection strategies. In foundational studies [30, 26, 32], candidate support triads $\{P_k, \vec{t}_k\}_{k=1}^{3}$ were constructed by referencing background spatial structures, such as macro-element planar subdivisions or Delaunay triangulations. While effective for canonical geometries, relying on underlying meshes contradicts the core "meshless" promise of unconstrained point-cloud discretization [10, 31].

When attempting to liberate the VNMM from spatial meshes by performing automated nearest-neighbor searches (e.g., k-d tree queries) over arbitrary unstructured point clouds, a severe mathematical limitation emerges: **the numerical approximation of the curl operator $(\nabla \times \vec{E}^h)_z$ fails to converge, exhibiting an asymptotic $\mathcal{O}(1)$ relative error plateau regardless of spatial grid refinement** $h \to 0$ [31].

Unlike Edge Finite Element Methods (EFEM), where boundary line circulation integrals enforce closed-contour flux cancellation via Stokes' theorem [7, 3], point-collocated meshless schemes rely on discrete pointwise projections [31]. In the classical 3-node solenoidal basis $\mathcal{L}^1$, polynomial incompleteness prevents local shape functions from resolving longitudinal (normal) field derivative components ($\partial E_x/\partial x$, $\partial E_y/\partial y$). Under pointwise collocation, unrepresented normal field gradients generate truncation residuals that interact with the inverted local moment placement matrix, leaking directly into the field curl calculation as an irreducible modal aliasing noise floor [34, 10].

A complete, step-by-step mathematical derivation of this modal aliasing mechanism—including the local Taylor series Jacobian decomposition and remainder error scaling—is presented in detail in Section 2.2.5.

1.3. Scope and Main Contributions of the Paper

This paper presents a stabilized Vector Nodal Meshless Method (VNMM 2D) for robust and accurate vector approximation on arbitrary point clouds without mesh connectivity. By resolving the mathematical origin of modal aliasing, establishing algebraic support qualification, and introducing conforming hybrid domain decomposition, this work delivers five primary contributions to computational electromagnetics:

Table 1: Conceptual comparison of spatial discretization paradigms in computational electromagnetics.

Discretization Method	**Mesh Connectivity**	**Field Variable Representation**	**$H(\mathbf{curl})$ Interface Continuity**	**Spurious-Mode Control Mechanism**	**Curl Operator Convergence on Point Clouds**
Edge FEM (EFEM) [7, 3]	Conformal mesh	Edge circulation $e_k = \int_{E_k} \vec{E} \cdot d\boldsymbol{\ell}$	Natively conforming ($\mathbf{n} \times [\vec{E}] = \mathbf{0}$)	Exact discrete de Rham complex	$\mathcal{O}(h^1)$ (simplex-based)
Scalar EFG / MLS [14, 21]	None (Point cloud)	Component splitting (E_x, E_y)	Artificial (C^0 normal locking / penalty)	Auxiliary divergence penalty or projection	$\mathcal{O}(h^p)$ (high-order scalar MLS)
Vector MLPG [13, 23]	None (Local subdomains)	Component splitting (E_x, E_y)	Lagrange multipliers / penalty	Divergence-free projection / penalty	$\mathcal{O}(h^p)$ (local MLS integration)
RBF-FD / RPIM [12, 11]	None (Point cloud)	Component splitting (E_x, E_y)	Collocation / ghost node penalty	Divergence cleaning / penalty	$\mathcal{O}(h^p)$ (radial basis differentiation)
Original VNMM ($\mathcal{L}^1$) [30, 26]	Planar / Delaunay subdivision	Pointwise projection $c_k = \vec{E}(P_k) \cdot \vec{t}_k$	Natural tangential director alignment	Conditional (restricted to structured supports)	$\mathcal{O}(1)$ stagnation on arbitrary clouds
Proposed VNMM ($\mathcal{P}^1$)	None (Arbitrary point clouds)	Pointwise projection $c_k = \vec{E}(P_k) \cdot \vec{t}_k$	Natural tangential director alignment	Div-curl Ritz-Galerkin penalty ($s_{\text{div}} = 6.0$)	Consistent $\mathcal{O}(h^1)$ via full Jacobian space

1. **Transition to Mesh-Connectivity-Free Support Qualification:** By replacing pre-defined spatial subdivisions with a 4-phase automated support selection algorithm combining k-d tree spatial queries with scale-dependent determinant thresholding, the proposed solver automatically identifies well-conditioned support sextets across arbitrary node clouds. As detailed in Section 3 and evaluated with random spatial jitter in Section 6.3, this removes background mesh connectivity requirements and enables robust operation under unconstrained vector orientations.
2. **Scale-Dependent Support Selection with Absolute Determinant Floor:** We introduce an algebraic singularity mode classification in $\mathbb{R}^{6\times 6}$ (partial vector parallelism, partial spatial collinearity, total 6-node collinearity, and azimuthal divergence masking), as presented in Sections 3.1 and 3.2. Enforcing an absolute tolerance floor within the scale-dependent determinant thresholding law (Section 3.3.1) provides a unified algebraic criterion for rejecting ill-conditioned local configurations, maintaining bounded matrix condition numbers ($\kappa(\mathbf{A}) \leq 73$, with values near ≈ 64 at the best-performing threshold) and restoring monotonic convergence under a representative 25% stochastic spatial jitter (Section 6.3).
3. **Elimination of Modal Aliasing via Complete $\mathcal{P}^1$ Vector Basis:** Transitioning from the incomplete 3-term solenoidal space $\mathcal{L}^1$ to the complete linear polynomial vector space $\mathcal{P}^1 = \mathcal{P}_1 \times \mathcal{P}_1$ resolves all four independent spatial components of the local Jacobian tensor $\mathbf{J}^h = \nabla \vec{E}^h$, as formulated in Section 2.3 following the aliasing diagnosis in Section 2.2.5. As demonstrated numerically in Section 6.2, this removes the derivative modal aliasing mechanism, breaking the $\mathcal{O}(1)$ curl stagnation plateau of $\mathcal{L}^1$ and restoring first-order convergence $\mathcal{O}(h^{0.97})$ for the curl operator $(\nabla \times \vec{E}^h)_z$ alongside an observed rate $\mathcal{O}(h^{2.17})$ for the electric field $\vec{E}^h$ (19.8× precision gain and 25.9× curl error reduction on the finest mesh).
4. **Spectral Purification via Div-Curl Regularization:** To suppress non-physical spurious gradient modes caused by discrete nullspace leakage ($\nabla\times(\nabla\phi) \neq \mathbf{0}$ at quadrature points), we implement a Ritz-Galerkin weak formulation regularized by a div-curl penalty term ($s_{\text{div}} = 6.0$). As developed in Section 4 and validated in Section 6.4, this shifts irrotational gradient states far outside the operational frequency window

($\lambda > 50$), yielding spurious-free spectral pairing across the investigated TE_z bandwidth and restricting the mean cutoff wavenumber error to 1.00%.

5. **Conforming Master-Slave Hybrid EFEM-VNMM Coupling:** Developed in Section 5, we establish a conforming domain decomposition strategy connecting Whitney edge finite elements and VNMM midpoint nodes through an exact dimensional scale transformation $c_k = e_k/l_k$. As validated in Section 6.5, this direct reduction eliminates interface DoFs without auxiliary Lagrange multipliers, avoiding zero-mass saddle-point indefiniteness, preserving symmetric positive-definite (SPD) system properties under regularized boundary conditions, and preventing spurious infinite eigenvalues ($\lambda = \infty$).

The remainder of this paper is organized as follows: Section 2 details the 2D VNMM vector shape function construction for $\mathcal{L}^1$ and $\mathcal{P}^1$ spaces and proves the Dual Equivalence Theorem; Section 3 analyzes local placement matrix singularities and formulates the support selection algorithm; Section 4 presents the div-curl regularized weak form and point-based EFG Gauss quadrature; Section 5 develops the conforming master-slave hybrid FEM-VNMM solver; Section 6 presents numerical benchmarks and convergence studies; and Section 7 concludes the paper with future research perspectives.

2. Mathematical Formulations of the 2D VNMM

This section establishes the foundational mathematical and functional framework of the two-dimensional Vector Nodal Meshless Method (VNMM 2D). It details the vector projection properties, field continuity enforcement, and the construction of local vector shape functions. First, the single-degree-of-freedom nodal projection scheme and the exact imposition of physical boundary conditions are defined. Next, the traditional incomplete $\mathcal{L}^1$ solenoidal space is reviewed, proving its exact topological and algebraic dual equivalence with classical Whitney 1-form edge finite elements. Finally, the complete linear polynomial vector space $\mathcal{P}^1$ is introduced to resolve all components of the local Taylor Jacobian tensor and eliminate derivative modal aliasing in unconstrained node clouds.

2.1. Nodal Projection and Tangential Continuity

The Vector Nodal Meshless Method (VNMM 2D) [30, 26] constructs vector approximations directly within $H(\mathbf{curl})$ Sobolev spaces by distributing a set of M nodes throughout the computational domain Ω and its boundaries $\partial\Omega$. Unlike traditional scalar meshless formulations that assign separate degrees of freedom to individual spatial components (e.g., E_x and E_y), the VNMM associates a single physical unit vector director $\vec{t_i} = [t_{ix}, t_{iy}]^T$ with each node i.

The approximate vector field $\vec{E}^h(\mathbf{x})$ at any spatial location $\mathbf{x} = (x, y) \in \Omega$ is expressed as a linear combination of vector nodal shape functions $\vec{N}_i(\mathbf{x})$:

$$\vec{E}^h(x, y) = \sum_{i=1}^{M_{\text{supp}}} \vec{N}_i(x, y) e_i \tag{1}$$

where M_{supp} is the number of active support nodes in the local influence domain, and e_i represents the discrete scalar degree of freedom (DoF) associated with the i-th node. Physically, e_i corresponds to the pointwise directional projection of the electric field vector along the unit vector director $\vec{t_i}$ at the node coordinate $\mathbf{x}_i = (x_i, y_i)$:

$$e_i = \vec{E}(\mathbf{x}_i) \cdot \vec{t_i} \quad [\text{Volts/meter}] \tag{2}$$

To ensure physical consistency, curl-conformity, and direct enforcement of boundary conditions, the vector shape functions $\vec{N}_i(\mathbf{x})$ are required to satisfy the vector Kronecker delta projection property [27, 30]:

$$\vec{N}_i(\mathbf{x}_k) \cdot \vec{t}_k = \delta_{ik} = \begin{cases} 1, & \text{if } k = i \\ 0, & \text{if } k \neq i \end{cases} \tag{3}$$

for all nodes k belonging to the local support domain. Property (3) dictates that the shape function $\vec{N}_i$ exhibits a unit projection along its own nodal director $\vec{t_i}$ at position $\mathbf{x}_i$, while maintaining zero projection along the directors of all other surrounding support nodes.

2.1.1. Enforcement of PEC and Material Interface Conditions

The vector Kronecker delta property (3) enables a straightforward, exact enforcement of essential Dirichlet boundary conditions and tangential field

continuity without requiring background auxiliary meshes, penalty parameters, or Lagrange multipliers [32, 30].

For nodes lying on Perfect Electric Conductor (PEC) boundaries Γ_{PEC}, where the physical boundary condition requires $\hat{n} \times \vec{E} = 0$ ($\hat{n}$ being the unit normal to $\partial\Omega$), the associated vector director $\vec{t_k}$ is aligned strictly tangent to the boundary contour:

$$\vec{t_k} \parallel \partial\Omega \implies \hat{n}(\mathbf{x}_k) \cdot \vec{t_k} = 0 \tag{4}$$

Consequently, the physical requirement $\hat{n} \times \vec{E} = 0$ simplifies to setting the scalar nodal DoF directly to zero:

$$e_k = 0 \quad \forall k \in \Gamma_{\text{PEC}} \tag{5}$$

Similarly, at material interfaces Γ_{int} separating subdomains with distinct constitutive parameters ($\epsilon_{r1} \neq \epsilon_{r2}$ or $\mu_{r1} \neq \mu_{r2}$), interface nodes are assigned unit vector directors $\vec{t_i}$ oriented tangentially to Γ_{int}. Because property (3) locks the tangential field component to e_i, tangential continuity ($\vec{n} \times \vec{E}_1 = \vec{n} \times \vec{E}_2$) is inherently satisfied across adjacent local support domains, while allowing natural step discontinuities in the normal field components ($\epsilon_{r1} E_{n1} = \epsilon_{r2} E_{n2}$).

2.2. The Incomplete $\mathcal{L}^1$ Basis (3 Support Nodes)

The construction of vector shape functions in the 2D VNMM directly inherits the mathematical foundations of $H(\mathbf{curl})$-conforming edge-based meshless approximations [27, 28]. The incomplete 3-term solenoidal polynomial space

$$\mathcal{L}^1 = \text{span}\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} y \\ -x \end{bmatrix} \} \tag{6}$$

corresponds to the local two-dimensional reduction of Nédélec's first-type space [7], whose vector projection mechanisms were established for edge meshless line segments by Silva et al. [28] and later adapted to point-collocated nodes [30, 26].

Consequently, the local vector shape function $\vec{N_i}$ associated with the i-th support node in a 3-node local domain is defined as a linear combination of these basis vectors:

$$\vec{N_i}(x, y) = \beta_{1i} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \beta_{2i} \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \beta_{3i} \begin{bmatrix} y \\ -x \end{bmatrix} \tag{7}$$

where $\boldsymbol{\beta}_i = [\beta_{1i}, \beta_{2i}, \beta_{3i}]^T \in \mathbb{R}^3$ represents the unknown local interpolation coefficient vector.

To determine these coefficients, the vector shape functions are constrained to satisfy the vector Kronecker delta projection property (3) for $i, k = 1, 2, 3$,

Applying constraint (3) to all three support nodes leads to the local 3×3 linear placement system:

$$\mathbf{A}\boldsymbol{\beta}_i = \mathbf{L}_i \tag{8}$$

where the local moment placement matrix $\mathbf{A} \in \mathbb{R}^{3\times3}$ is explicitly given by:

$$\mathbf{A} = \begin{bmatrix} t_{1x} & t_{1y} & y_1 t_{1x} - x_1 t_{1y} \\ t_{2x} & t_{2y} & y_2 t_{2x} - x_2 t_{2y} \\ t_{3x} & t_{3y} & y_3 t_{3x} - x_3 t_{3y} \end{bmatrix} \tag{9}$$

and the canonical load vectors are $\mathbf{L}_1 = [1, 0, 0]^T$, $\mathbf{L}_2 = [0, 1, 0]^T$, and $\mathbf{L}_3 = [0, 0, 1]^T$.

Once the coefficients $\boldsymbol{\beta}_i = \mathbf{A}^{-1}\mathbf{L}_i$ are evaluated, the approximated electric field vector $\vec{E}^h(x, y)$ is constructed via nodal projection superposition:

$$\vec{E}^h(x, y) = \sum_{i=1}^{3} \vec{N}_i(x, y) e_i \tag{10}$$

where $e_i = \vec{E}(\mathbf{x}_i) \cdot \vec{t}_i$ represent the discrete scalar projection DoFs.

2.2.1. Mathematical Properties and Solenoidal Identity

The shape functions $\vec{N}_i$ of the $\mathcal{L}^1$ basis possess two distinct mathematical characteristics:

1. **Identical Solenoidal Property ($\nabla \cdot \vec{N}_i \equiv 0$):** Taking the divergence of Eq. (7) yields:

$$\nabla \cdot \vec{N}_i = \frac{\partial}{\partial x}(\beta_{1i} + \beta_{3i} y) + \frac{\partial}{\partial y}(\beta_{2i} - \beta_{3i} x) \equiv 0 \tag{11}$$

2. **Univariate Curl Representation:** Evaluating the scalar z-directed curl operator on $\vec{N}_i$ yields:

$$(\nabla \times \vec{N}_i)_z = \frac{\partial N_{iy}}{\partial x} - \frac{\partial N_{ix}}{\partial y} = \frac{\partial(-\beta_{3i} x)}{\partial x} - \frac{\partial(\beta_{3i} y)}{\partial y} = -2\beta_{3i} \tag{12}$$

Therefore, the approximated z-directed scalar curl field $(\nabla \times \vec{E}^h)_z$ is spatially constant across the 3-node support domain and depends exclusively on the third coefficient β_{3i}:

$$(\nabla \times \vec{E}^h)_z = -2 \sum_{i=1}^{3} \beta_{3i} e_i \tag{13}$$

2.2.2. Dual Whitney Equivalence and Constructive Isomorphism

The VNMM 2D was originally conceptualized as the zero-edge-length asymptotic limit of the Edge Meshless Method (EMM) [27, 30]. By examining this heritage, a fundamental mathematical duality and constructive isomorphism between the VNMM point-collocation formulation and classic lowest-order Whitney edge finite elements (Nédélec 1-forms on triangular simplices [7, 3]) is formally established, as detailed in Fig. 1.

In conventional FEM (Fig. 1a), the degrees of freedom associated with the triangular cell Ω_e are line integrals of circulation $\alpha_k = \oint_{e_k} \vec{E} \cdot d\boldsymbol{\ell}$ along continuous 1D edges of finite physical length L_k. As demonstrated in the asymptotic collapse limit (Fig. 1b), when the physical edge length shrinks to zero ($L_k \to 0$), the 1D continuous edge degenerates into a point-collocated vector node $(P_k, \vec{t}_k)$, where the circulation DoF α_k naturally scales into a pointwise directional field projection $e_k = \vec{E}(P_k) \cdot \vec{t}_k$ [Volts/meter].

2.2.3. Formal Proof of the Dual Equivalence Theorem

Let $T \subset \mathbb{R}^2$ be an arbitrary non-degenerate triangular simplex with vertices V_1, V_2, V_3 ordered counter-clockwise and non-zero area $|T| > 0$. Let $e_k = (V_k, V_{k+1})$ be the directed boundary edges of length $L_k = \|V_{k+1} - V_k\| > 0$, with unit tangent vectors $\vec{t}_k = (V_{k+1} - V_k)/L_k$ and midpoints $M_k = \frac{1}{2}(V_k + V_{k+1})$ for $k \in \{1, 2, 3\}$ (with index wrap-around $V_4 \equiv V_1$).

Theorem 1 (Dual Equivalence of Whitney 1-Forms and VNMM 2D $\mathcal{L}^1$)**.** *Consider the support triad of vector nodes $\{P_k, \vec{t}_k\}_{k=1}^3$ defined at the edge midpoints $P_k = M_k$ with directors $\vec{t}_k$. The VNMM 2D $\mathcal{L}^1$ shape functions $\vec{N}_k(\mathbf{x})$ and the Whitney 1-forms $\vec{w}_k(\mathbf{x}) = \lambda_i(\mathbf{x})\nabla\lambda_j - \lambda_j(\mathbf{x})\nabla\lambda_i$ satisfy:*

$$\vec{N}_k(\mathbf{x}) \equiv L_k \, \vec{w}_k(\mathbf{x}) \quad \Longleftrightarrow \quad \vec{w}_k(\mathbf{x}) \equiv \frac{1}{L_k} \vec{N}_k(\mathbf{x}), \quad \forall \mathbf{x} \in \mathbb{R}^2, \ \forall k \in \{1, 2, 3\} \tag{14}$$

Proof. The proof proceeds in five constructive steps:

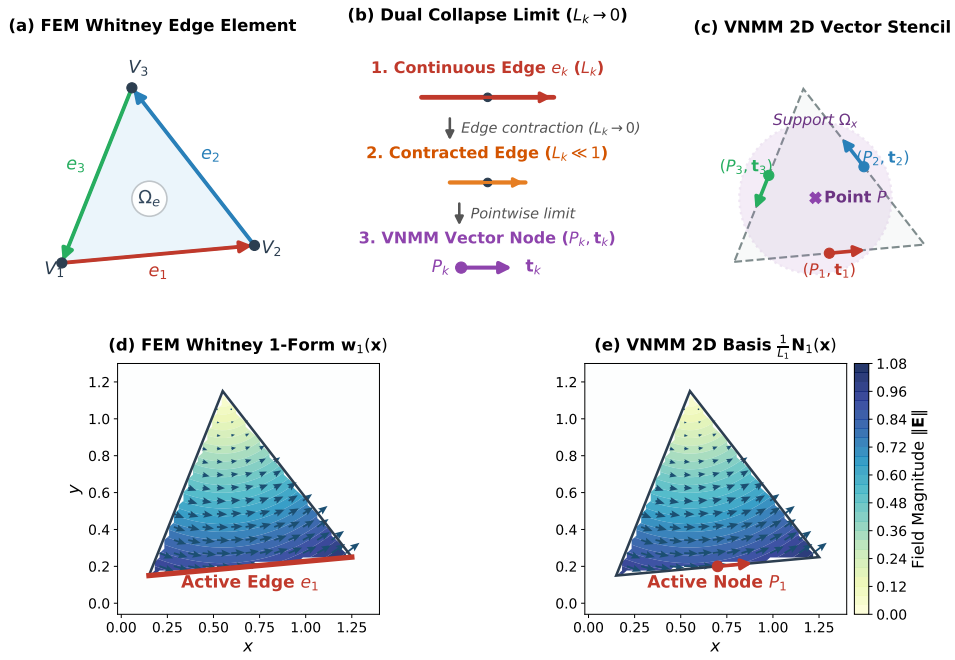


Figure 1: Exact topological and algebraic dual equivalence between lowest-order Whitney edge elements (FEM 1-forms) and the VNMM 2D $\mathcal{L}^1$ meshless vector formulation: (a) Primal triangular cell Ω_e with vertices V_1, V_2, V_3 and oriented edges e_1, e_2, e_3; (b) Conceptual dual collapse limit as edge length contracts ($L_k \to 0$), identifying the discrete circulation integral $L_k^{-1} \int_{e_k} \vec{E} \cdot \vec{t}_k \, ds$ directly with the tangential projection at a point, $e_k = \vec{E}(P_k) \cdot \vec{t}_k$; (c) VNMM 2D meshless support stencil with vector nodes $(P_k, \vec{t}_k)$ placed at the edge midpoints and compact support Ω_x; (d) Vector field and magnitude contour of the Whitney 1-form $\vec{w}_1(\mathbf{x}) = \lambda_1 \nabla\lambda_2 - \lambda_2 \nabla\lambda_1$ associated with active edge e_1; (e) Vector field and magnitude contour of the normalized VNMM $\mathcal{L}^1$ vector basis function $\frac{1}{L_1}\vec{N}_1(\mathbf{x})$ associated with active node $(P_1, \vec{t}_1)$. Both fields are algebraically identical over Ω_e, satisfying $\|\vec{w}_1(\mathbf{x}) - L_1^{-1}\vec{N}_1(\mathbf{x})\|_{L^\infty(\Omega_e)} \leq 2.2 \times 10^{-16}$.

Step 1: Isomorphism of Approximation Spaces. The lowest-order Whitney space $\mathcal{W}(T)$ on T is spanned by $\vec{w}_k(\mathbf{x}) = [a_1 + \alpha y, a_2 - \alpha x]^T$. By inspection of the spanning vectors, $\mathcal{W}(T) = \text{span}\{[1,0]^T, [0,1]^T, [y,-x]^T\} \equiv \mathcal{L}^1$. Thus, both approximation spaces share identical 3-dimensional vector polynomial spaces.

Step 2: Constancy of Tangential Field Projection Along Simplex Edges. Let $\vec{u}(\mathbf{x}) = [a_1 + \alpha y, a_2 - \alpha x]^T \in \mathcal{L}^1$. The directional projection of

$\vec{u}(\mathbf{x})$ along unit director $\vec{t}_k = [\Delta x_k/L_k, \Delta y_k/L_k]^T$ is:

$$\vec{u}(\mathbf{x}) \cdot \vec{t}_k = \frac{1}{L_k} \left[a_1 \Delta x_k + a_2 \Delta y_k + \alpha(y\Delta x_k - x\Delta y_k) \right] \tag{15}$$

Since any point $\mathbf{x} = (x, y)$ on edge e_k satisfies $(x - x_k)\Delta y_k - (y - y_k)\Delta x_k = 0$, the rotational term $y\Delta x_k - x\Delta y_k = y_k\Delta x_k - x_k\Delta y_k$ is strictly constant along the entire edge length. Therefore, the tangential projection is invariant along e_k:

$$\vec{u}(\mathbf{x}) \cdot \vec{t}_k = \vec{u}(M_k) \cdot \vec{t}_k, \quad \forall \mathbf{x} \in e_k \tag{16}$$

Step 3: Reduction of Edge Circulation to Pointwise Midpoint Value. Using Eq. (16), the line integral defining the FEM circulation degree of freedom along edge e_j simplifies to:

$$\oint_{e_j} \vec{u} \cdot d\boldsymbol{\ell} = \int_{-L_j/2}^{L_j/2} \left(\vec{u}(\mathbf{x}(s)) \cdot \vec{t}_j \right) ds = \left(\vec{u}(M_j) \cdot \vec{t}_j \right) \int_{-L_j/2}^{L_j/2} ds = L_j \left(\vec{u}(M_j) \cdot \vec{t}_j \right) \tag{17}$$

Substituting the k-th Whitney basis function $\vec{w}_k \in \mathcal{W}(T)$, which satisfies $\oint_{e_j} \vec{w}_k \cdot d\boldsymbol{\ell} = \delta_{jk}$ by construction, yields:

$$L_j \left(\vec{w}_k(M_j) \cdot \vec{t}_j \right) = \delta_{jk} \quad \Longrightarrow \quad (L_k \vec{w}_k(M_j)) \cdot \vec{t}_j = \delta_{jk} \tag{18}$$

Step 4: Uniqueness and Exact Identification with VNMM Shape Functions. By definition of the VNMM 2D $\mathcal{L}^1$ interpolation problem on the support triad $\{M_j, \vec{t}_j\}_{j=1}^3$, the shape functions satisfy $\vec{N}_k(M_j) \cdot \vec{t}_j = \delta_{jk}$. Consider the difference field:

$$\vec{d}_k(\mathbf{x}) = \vec{N}_k(\mathbf{x}) - L_k \vec{w}_k(\mathbf{x}) \in \mathcal{L}^1 \tag{19}$$

Evaluating its projection at each midpoint M_j:

$$\vec{d}_k(M_j) \cdot \vec{t}_j = \vec{N}_k(M_j) \cdot \vec{t}_j - L_k \left(\vec{w}_k(M_j) \cdot \vec{t}_j \right) = \delta_{jk} - \delta_{jk} = 0, \quad \forall j \in \{1, 2, 3\} \tag{20}$$

Because triangle T is non-degenerate, the local moment placement matrix $\mathbf{A} \in \mathbb{R}^{3\times3}$ is non-singular ($\det \mathbf{A} \neq 0$). The only vector field in $\mathcal{L}^1$ satisfying zero projections across all three support directors is the trivial field $\vec{d}_k(\mathbf{x}) \equiv \mathbf{0}$, which proves the exact identity $\vec{N}_k(\mathbf{x}) \equiv L_k \vec{w}_k(\mathbf{x})$ everywhere in $\mathbb{R}^2$.

Step 5: Verification of Differential Invariants. 1. *Divergence:* For any field $\vec{u} = [a_1 + \alpha y, a_2 - \alpha x]^T \in \mathcal{L}^1$, we have $\nabla \cdot \vec{u} = \partial(a_1 + \alpha y)/\partial x +$

$\partial(a_2 - \alpha x)/\partial y \equiv 0$. Thus, $\nabla \cdot \vec{w}_k \equiv 0$ and $\nabla \cdot \vec{N}_k \equiv 0$ everywhere. 2. *Curl:* Applying Stokes' theorem over triangle T for Whitney 1-form $\vec{w}_k$:

$$\int_T (\nabla \times \vec{w}_k)_z d\Omega = \oint_{\partial T} \vec{w}_k \cdot d\boldsymbol{\ell} = \sum_{j=1}^{3} \delta_{jk} = 1 \implies (\nabla \times \vec{w}_k)_z = \frac{1}{|T|} \quad (21)$$

By the scaling law (14), $(\nabla \times \vec{N}_k)_z = L_k(\nabla \times \vec{w}_k)_z = L_k/|T|$. □

This constructive isomorphism is graphically presented in Fig. 1d and 1e and is verified numerically in Section 6.1.

2.2.4. Extension of the Valid Physical Interpolation Domain

This dual Whitney equivalence resolves a longstanding misconception regarding meshless vector interpolation domains [31]. In conventional scalar meshless methods (e.g., Moving Least Squares or Element-Free Galerkin), evaluating field approximations outside the physical convex hull enclosed by the node coordinates $\mathbf{x}_k$ constitutes an unconstrained extrapolation, leading to rapid truncation error growth [27].

However, because the VNMM vector basis reproduces the complete polynomial space of Whitney edge elements over the virtual macro-triangle Ω_x defined by the intersecting dotted lines of action of vector directors $\vec{t}_k$ (Fig. 1c), the valid domain for $H(\mathbf{curl})$ field approximation, tangential field continuity, and constant curl representation is not geographically restricted to the small physical convex hull of $\mathbf{x}_k$ [7, 3]. Instead, as proven by the dual scaling law (14), it extends throughout the entire interior bounding region of the virtual macro-triangle.

As long as the support selection algorithm enforces a well-conditioned matrix $\mathbf{A}$ (guaranteeing that the intersection vertices remain bounded and non-degenerate), field evaluations at any point $P \in \Omega_x$ preserve $H(\mathbf{curl})$ conformity and satisfy standard Bramble-Hilbert interpolation bounds $\mathcal{O}(h)$ for fields and $\mathcal{O}(1)$ or $\mathcal{O}(h)$ for curl operators depending on the basis completeness [3].

2.2.5. Mathematical Origin of Modal Aliasing and Curl Stagnation

Despite its solenoidal identity and dual equivalence to Whitney edge elements, the incomplete $\mathcal{L}^1$ basis suffers from a fundamental mathematical limitation when applied to point-collocated meshless approximations on arbitrary point clouds. Specifically, the numerical approximation of the curl

operator $(\nabla \times \vec{E}^h)_z$ fails to achieve asymptotic convergence, stagnating at an irreducible $\mathcal{O}(1)$ relative error plateau regardless of spatial grid refinement $h \to 0$ [31].

The mathematical origin of this derivative stagnation lies in the local Taylor series expansion of the exact electric field $\vec{E}$ centered at the evaluation point $P(x_P, y_P)$ [34, 10]:

$$\vec{E}(\mathbf{x}) = \vec{E}(P) + \mathbf{J}(P) \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} + \mathcal{O}(h^2) \tag{22}$$

where $\mathbf{J}(P) = \nabla \vec{E}$ represents the complete first-order field Jacobian tensor [10]:

$$\mathbf{J} = \begin{bmatrix} \frac{\partial E_x}{\partial x} & \frac{\partial E_x}{\partial y} \\ \frac{\partial E_y}{\partial x} & \frac{\partial E_y}{\partial y} \end{bmatrix} \tag{23}$$

By expressing individual spatial derivatives in terms of fundamental vector calculus operators—namely, the field divergence $\nabla \cdot \vec{E} = \partial E_x/\partial x + \partial E_y/\partial y$ and the z-directed field curl $(\nabla \times \vec{E})_z = \partial E_y/\partial x - \partial E_x/\partial y$—the field Jacobian tensor is decomposed as:

$$\mathbf{J} = \frac{1}{2}(\nabla \cdot \vec{E})\mathbf{I} + \frac{1}{2}(\nabla \times \vec{E})_z \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} + \begin{bmatrix} \delta_{\text{norm}} & -\delta_{\text{trans}} \\ \delta_{\text{trans}} & -\delta_{\text{norm}} \end{bmatrix} \tag{24}$$

where $\delta_{\text{norm}} = \frac{1}{2}(\partial E_x/\partial x - \partial E_y/\partial y)$ denotes the normal (longitudinal) field derivative difference, and $\delta_{\text{trans}} = \frac{1}{2}(\partial E_y/\partial x + \partial E_x/\partial y)$ represents the symmetric transverse field derivative component.

In the incomplete $\mathcal{L}^1$ basis, local shape functions provide only 3 degrees of freedom. As a result, normal field derivatives are artificially forced to vanish identically ($\partial E_x^h/\partial x \equiv 0$ and $\partial E_y^h/\partial y \equiv 0$), imposing $\nabla \cdot \vec{E}^h \equiv 0$ and $\delta_{\text{norm}} \equiv 0$ simultaneously.

This fundamental limitation is formalized in the following proposition:

Proposition 1 (Jacobian Incompleteness and Derivative Modal Aliasing)**.** *Let $\vec{E} \in [C^2(\Omega)]^2$ be an arbitrary twice continuously differentiable 2D vector field. The incomplete solenoidal space $\mathcal{L}^1 = \text{span}\{[1,0]^T, [0,1]^T, [-y,x]^T\} \subset \mathcal{P}_1 \times \mathcal{P}_1$ cannot reproduce the complete first-order Taylor expansion of $\vec{E}$. Specifically, while the complete space $\mathcal{P}^1 = \mathcal{P}_1 \times \mathcal{P}_1$ possesses $\dim(\mathcal{P}^1) = 6$ degrees of freedom to reconstruct the 2 translation components and all 4 independent entries of the Jacobian tensor $\mathbf{J} = \nabla \vec{E}$, $\mathcal{L}^1$ possesses only $\dim(\mathcal{L}^1) = 3$*

degrees of freedom, enforcing the artificial linear constraints:

$$\frac{\partial E_x^h}{\partial x} = 0, \quad \frac{\partial E_y^h}{\partial y} = 0, \quad \textit{and} \quad \frac{\partial E_x^h}{\partial y} + \frac{\partial E_y^h}{\partial x} = 0. \tag{25}$$

Consequently, under point-collocated projection on unstructured node clouds, unrepresented normal gradients ($\partial E_x/\partial x, \partial E_y/\partial y \neq 0$) generate a projection remainder $\mathbf{r} \sim \mathcal{O}(h)$ that couples into the rotational coefficient through the inverted moment placement matrix $\mathbf{A}^{-1} \sim \mathcal{O}(h^{-1})$, inducing an irreducible $\mathcal{O}(1)$ derivative aliasing error in the discrete curl operator:

$$\|(\nabla \times \vec{E})_z - (\nabla \times \vec{E}^h)_z\|_{L^2(\Omega)} \sim \|\mathbf{A}^{-1}\mathbf{r}\| = \mathcal{O}(h^{-1}) \cdot \mathcal{O}(h) = \mathcal{O}(1). \tag{26}$$

Proof. Let any discrete vector field $\vec{E}^h \in \mathcal{L}^1$ be expanded in terms of the solenoidal basis:

$$\vec{E}^h(x, y) = a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \end{bmatrix} + c \begin{bmatrix} -y \\ x \end{bmatrix}, \quad a, b, c \in \mathbb{R}. \tag{27}$$

Direct differentiation with respect to spatial coordinates yields the discrete Jacobian tensor:

$$\mathbf{J}^h = \begin{bmatrix} \frac{\partial E_x^h}{\partial x} & \frac{\partial E_x^h}{\partial y} \\ \frac{\partial E_y^h}{\partial x} & \frac{\partial E_y^h}{\partial y} \end{bmatrix} = \begin{bmatrix} 0 & -c \\ c & 0 \end{bmatrix}. \tag{28}$$

Inspecting Eq. (28) establishes that $\partial E_x^h/\partial x \equiv 0$, $\partial E_y^h/\partial y \equiv 0$, and $\partial E_x^h/\partial y + \partial E_y^h/\partial x = -c + c \equiv 0$, identically satisfying the linear constraints in Eq. (25).

Now consider a general smooth physical field $\vec{E} \in [C^2(\Omega)]^2$. Its Taylor expansion centered at evaluation point P is $\vec{E}(\mathbf{x}) = \vec{E}(P) + \mathbf{J}(P)\Delta\mathbf{x} + \mathcal{O}(h^2)$, where $\mathbf{J}(P)$ has non-zero diagonal entries ($\partial E_x/\partial x \neq 0$ or $\partial E_y/\partial y \neq 0$). Collocating $\vec{E}^h$ at support nodes $\mathbf{x}_k$ ($k = 1, 2, 3$) along unit directors $\vec{t}_k$ defines the linear placement system $\mathbf{A}[a, b, c]^T = \mathbf{e}$, where $e_k = \vec{E}(\mathbf{x}_k) \cdot \vec{t}_k$.

Subtracting the projection of the unrepresented linear components $(\mathbf{J}(P) - \mathbf{J}^h)\Delta\mathbf{x}_k \cdot \vec{t}_k$ yields a nodal projection residual $\mathbf{r} \in \mathbb{R}^3$ with asymptotic magnitude $\|\mathbf{r}\| \sim \mathcal{O}(h)$. In the 3×3 moment placement matrix $\mathbf{A}$ (Eq. (9)), translation columns 1–2 scale as $\mathcal{O}(1)$, while the rotational moment column 3 scales as $(\Delta y_k t_{kx} - \Delta x_k t_{ky}) \sim \mathcal{O}(h)$. Consequently, by Cramer's rule or Neumann inversion, the third row of $\mathbf{A}^{-1}$ (which maps nodal data to the rotational DoF $c = -\frac{1}{2}(\nabla \times \vec{E}^h)_z$) scales strictly as $\| \operatorname{Row}_3(\mathbf{A}^{-1})\| \sim \mathcal{O}(h^{-1})$.

The numerical error in the reconstructed curl is therefore governed by the propagation of the projection remainder:

$$|(\nabla\times\vec{E})_z-(\nabla\times\vec{E}^h)_z| = 2|c_{\text{exact}}-c| \sim \|\operatorname{Row}_3(\mathbf{A}^{-1})\|\|\mathbf{r}\| = \mathcal{O}(h^{-1})\cdot\mathcal{O}(h) = \mathcal{O}(1), \tag{29}$$

which is independent of h and generates the irreducible $\mathcal{O}(1)$ stagnation plateau, completing the proof. □

Table 2: Comparison of polynomial completeness, dimensionality, and derivative reproduction capabilities between the incomplete solenoidal space $\mathcal{L}^1$ and the complete linear vector polynomial space $\mathcal{P}^1$.

Vector Space	dim	E_x, E_y	$\partial E_x/\partial x$	$\partial E_x/\partial y$	$\partial E_y/\partial x$	$\partial E_y/\partial y$
$\mathcal{L}^1$ (Incomplete solenoidal)	3	✓	×	✓	✓	×
$\mathcal{P}^1 = \mathcal{P}_1 \times \mathcal{P}_1$ (Complete linear)	6	✓	✓	✓	✓	✓

In Edge Finite Element Methods (EFEM), this lack of explicit normal derivative representation does not cause stagnation because continuous line circulation integrals along element edges enforce closed-contour flux cancellation via Stokes' theorem [7, 3]. In the VNMM, however, continuous line integrals are replaced by discrete pointwise projections [31]. Consequently, the $\mathcal{O}(1)$ derivative noise floor established in Proposition 1 cannot be eliminated by spatial mesh refinement ($h \to 0$), making the complete linear polynomial vector space $\mathcal{P}^1$ mathematically indispensable for unconstrained point clouds, as formulated in Section 2.3.

2.3. The Complete $\mathcal{P}^1$ Vector Polynomial Basis

To eliminate the derivative modal aliasing and curl stagnation inherent to the incomplete $\mathcal{L}^1$ space, we reformulate local vector interpolation using the complete linear polynomial vector space $\mathcal{P}^1 = \mathcal{P}_1 \times \mathcal{P}_1$. This transition draws a direct mathematical analogy to Nédélec's second family of mixed finite elements [35], in which vector field approximations are constructed over complete polynomial spaces $(\mathbb{P}_k)^d$ rather than solenoidal-restricted subspaces.

In two dimensions ($d = 2$), a complete first-order vector polynomial basis $\mathcal{P}^1$ provides 6 independent degrees of freedom ($M_{\text{supp}} = 6$ support nodes per evaluation point):

$$\mathcal{P}^1 = \operatorname{span}\left\{ \begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}, \begin{bmatrix}x\\0\end{bmatrix}, \begin{bmatrix}y\\0\end{bmatrix}, \begin{bmatrix}0\\x\end{bmatrix}, \begin{bmatrix}0\\y\end{bmatrix} \right\} \tag{30}$$

Using this complete basis, the local vector shape function $\vec{N}_i(\mathbf{x})$ associated with the i-th support node is expressed as a linear combination of its six constituent terms:

$$\vec{N}_i(x,y) = \beta_{1i}\begin{bmatrix}1\\0\end{bmatrix} + \beta_{2i}\begin{bmatrix}0\\1\end{bmatrix} + \beta_{3i}\begin{bmatrix}x\\0\end{bmatrix} + \beta_{4i}\begin{bmatrix}y\\0\end{bmatrix} + \beta_{5i}\begin{bmatrix}0\\x\end{bmatrix} + \beta_{6i}\begin{bmatrix}0\\y\end{bmatrix} \tag{31}$$

where β_{ji} (for $j = 1, \ldots, 6$) represent the local interpolation coefficients to be determined.

Enforcing the vector Kronecker delta projection property [30]:

$$\vec{N}_i(\mathbf{x}_k) \cdot \vec{t}_k = \delta_{ik}, \quad \text{for } k = 1, \ldots, 6 \tag{32}$$

across all six support nodes yields the local linear placement system

$$\mathbf{A}\boldsymbol{\beta}_i = \mathbf{L}_i. \tag{33}$$

Evaluated in local translated coordinates centered at the evaluation point $P(x_P, y_P)$ (defined as $\Delta x_k = x_k - x_P$ and $\Delta y_k = y_k - y_P$), the local moment placement matrix $\mathbf{A} \in \mathbb{R}^{6\times 6}$ is explicitly given by:

$$\mathbf{A} = \begin{bmatrix} t_{1x} & t_{1y} & \Delta x_1 t_{1x} & \Delta y_1 t_{1x} & \Delta x_1 t_{1y} & \Delta y_1 t_{1y} \\ t_{2x} & t_{2y} & \Delta x_2 t_{2x} & \Delta y_2 t_{2x} & \Delta x_2 t_{2y} & \Delta y_2 t_{2y} \\ t_{3x} & t_{3y} & \Delta x_3 t_{3x} & \Delta y_3 t_{3x} & \Delta x_3 t_{3y} & \Delta y_3 t_{3y} \\ t_{4x} & t_{4y} & \Delta x_4 t_{4x} & \Delta y_4 t_{4x} & \Delta x_4 t_{4y} & \Delta y_4 t_{4y} \\ t_{5x} & t_{5y} & \Delta x_5 t_{5x} & \Delta y_5 t_{5x} & \Delta x_5 t_{5y} & \Delta y_5 t_{5y} \\ t_{6x} & t_{6y} & \Delta x_6 t_{6x} & \Delta y_6 t_{6x} & \Delta x_6 t_{6y} & \Delta y_6 t_{6y} \end{bmatrix} \tag{34}$$

The corresponding coefficient vector $\boldsymbol{\beta}_i$ and canonical right-hand side vector $\mathbf{L}_i$ are structured as:

$$\boldsymbol{\beta}_i = \begin{bmatrix}\beta_{1i}\\ \beta_{2i}\\ \beta_{3i}\\ \beta_{4i}\\ \beta_{5i}\\ \beta_{6i}\end{bmatrix}, \qquad \mathbf{L}_i = \begin{bmatrix}\delta_{1i}\\ \delta_{2i}\\ \delta_{3i}\\ \delta_{4i}\\ \delta_{5i}\\ \delta_{6i}\end{bmatrix} \tag{35}$$

Solving the linear system (33) via $\boldsymbol{\beta}_i = \mathbf{A}^{-1}\mathbf{L}_i$ completely defines the vector shape functions $\vec{N}_i(\mathbf{x})$. Consequently, the approximated electric field

$\vec{E}^h(\mathbf{x})$ and its z-directed curl $(\nabla \times \vec{E}^h)_z$ within the support domain are expressed analytically as:

$$\vec{E}^h(x,y) = \sum_{i=1}^{6} \vec{N}_i(x,y) e_i \tag{36}$$

$$(\nabla \times \vec{E}^h)_z(x,y) = \frac{\partial E_y^h}{\partial x} - \frac{\partial E_x^h}{\partial y} = \sum_{i=1}^{6} (\beta_{5i} - \beta_{4i}) e_i \tag{37}$$

where $e_i = \vec{E}(P_i) \cdot \vec{t}_i$ denotes the scalar directional projection degree of freedom associated with the i-th node [30, 31].

By providing 6 independent degrees of freedom, the complete $\mathcal{P}^1$ space resolves all four spatial derivative components of the local Jacobian tensor $\mathbf{J}^h = \nabla \vec{E}^h$ simultaneously [35, 31]:

$$\mathbf{J}^h = \begin{bmatrix} \frac{\partial E_x^h}{\partial x} & \frac{\partial E_x^h}{\partial y} \\ \frac{\partial E_y^h}{\partial x} & \frac{\partial E_y^h}{\partial y} \end{bmatrix} = \sum_{i=1}^{6} \begin{bmatrix} \beta_{3i} & \beta_{4i} \\ \beta_{5i} & \beta_{6i} \end{bmatrix} e_i \tag{38}$$

Unlike the incomplete $\mathcal{L}^1$ space, which artificially forces longitudinal (normal) field derivatives ($\partial E_x/\partial x$, $\partial E_y/\partial y$) to vanish, the complete $\mathcal{P}^1$ basis resolves both rotational and irrotational (longitudinal flux variation) components simultaneously. This shields the field curl calculation from longitudinal derivative truncation residues, removing the derivative modal aliasing mechanism, restoring first-order asymptotic convergence $\mathcal{O}(h^1)$ for the curl operator $(\nabla \times \vec{E}^h)_z$, and providing quadratic convergence $\mathcal{O}(h^2)$ for electric field interpolation.

3. Algebraic Conditioning and Robust Support Selection

The construction of vector nodal shape functions $\vec{N}_i$ in the Vector Nodal Meshless Method (VNMM 2D) [30, 26] relies on solving a local linear placement system $\mathbf{A}\boldsymbol{\beta}_i = \mathbf{L}_i$. A fundamental prerequisite for the existence, uniqueness, and numerical stability of these vector shape functions is that the local moment placement matrix $\mathbf{A}$ remains strictly non-singular ($|\det(\mathbf{A})| > 0$).

Unlike scalar meshless formulations where matrix singularity is dictated purely by spatial nodal distributions (e.g., node co-location or spatial co-linearity) [27], the conditioning of $\mathbf{A}$ in the VNMM is intrinsically coupled

to both the spatial coordinates $\mathbf{x}_k = (x_k, y_k)$ and the orientation of the unit vector directors $\vec{t}_k = [t_{kx}, t_{ky}]^T$ assigned to each support node [30, 31]. A comprehensive classification of these mathematical breakdown modes (Figs. 2a-2d), the spatial-directional decoupling principle (Fig. 2e), and the proposed robust support qualification (Fig. 2f) is illustrated in Fig. 2.

3.1. Singularities and Geometric Constraints in the Incomplete $\mathcal{L}^1$ Basis (3×3 Matrix)

To establish a systematic classification of algebraic singularities in the original VNMM 2D formulation [30, 26], we revisit the 3×3 local moment placement matrix $\mathbf{A}$ generated by the 3-node solenoidal basis $\mathcal{L}^1$. Originally derived in Eq. (9), $\mathbf{A}$ is restated below to facilitate the singularity analysis:

$$\mathbf{A} = \begin{bmatrix} t_{1x} & t_{1y} & y_1 t_{1x} - x_1 t_{1y} \\ t_{2x} & t_{2y} & y_2 t_{2x} - x_2 t_{2y} \\ t_{3x} & t_{3y} & y_3 t_{3x} - x_3 t_{3y} \end{bmatrix} \tag{39}$$

For this matrix, algebraic singularity ($\det(\mathbf{A}) = 0$) occurs when the spatial-directional equations contributed by the candidate triad become linearly dependent [26, 31]. As classified visually in Figs. 2a–2d, these mathematical failures fall into four distinct pathological configurations:

1. **Global Vector Parallelism (Fig. 2a):** If all three unit directors are parallel or anti-parallel ($\vec{t}_1 \parallel \vec{t}_2 \parallel \vec{t}_3$), director components satisfy $t_{kx} = c \cdot t_{ky}$ for a constant scalar c. Thus, $\mathrm{Col}_1(\mathbf{A}) = c \cdot \mathrm{Col}_2(\mathbf{A})$, yielding $\det(\mathbf{A}) \equiv 0$ regardless of nodal spatial distribution.
2. **Concurrency of Lines of Action (Fig. 2b):** If the infinite lines of action passing through $\mathbf{x}_k$ along $\vec{t}_k$ intersect at a single common point $P_c(x_c, y_c)$, every node satisfies $t_{kx}(y_k - y_c) - t_{ky}(x_k - x_c) = 0$. This forces the rotational column Col_3 to become a linear combination of translation columns, $\mathrm{Col}_3 = y_c\,\mathrm{Col}_1 - x_c\,\mathrm{Col}_2$, causing $\det(\mathbf{A}) = 0$.
3. **Overlapping Lines of Action (Fig. 2c):** If two support nodes lie on a common line $\mathcal{L}$ with directors aligned parallel to $\mathcal{L}$, the second row becomes identical to the first ($\mathrm{Row}_2 = \mathrm{Row}_1$), causing rank deficiency ($\mathrm{rank}(\mathbf{A}) \leq 2$).
4. **Nodal Coincidence / Spatial Collapse (Fig. 2d):** Shared spatial coordinates ($\mathbf{x}_i \approx \mathbf{x}_j$) collapse spatial moment sampling, preventing 2D vector interpolation span.

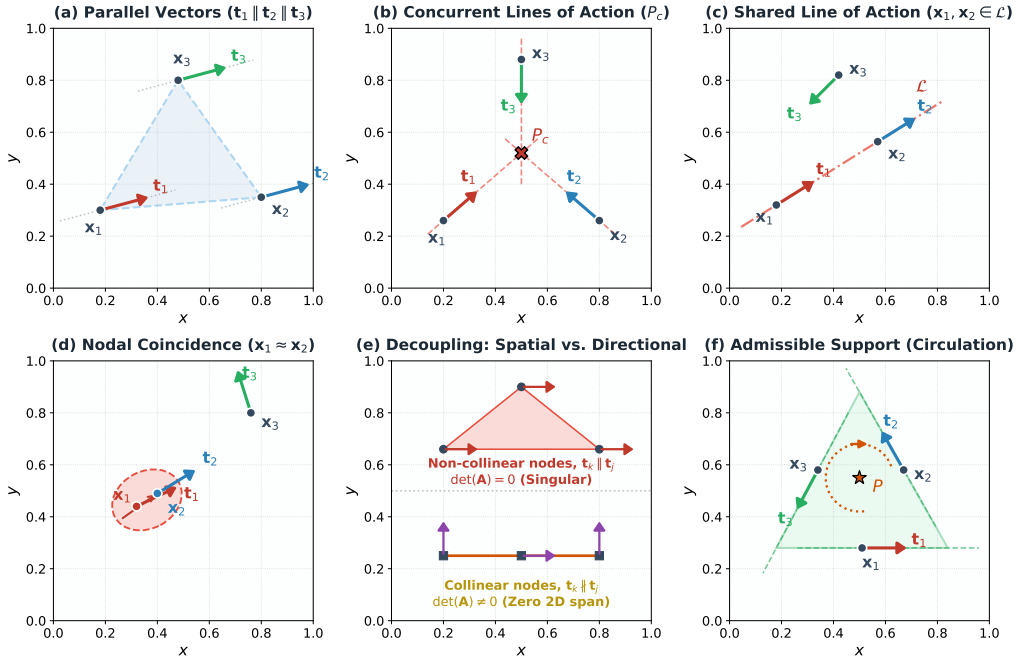


Figure 2: Geometric and directional singularity configurations of the VNMM 2D local placement matrix $\mathbf{A} \in \mathbb{R}^{3\times 3}$ for the $\mathcal{L}^1$ solenoidal vector basis: (a) **Global vector parallelism** ($\vec{t}_1 \parallel \vec{t}_2 \parallel \vec{t}_3$): yields $t_{kx} = c\, t_{ky}$ for all k, causing linear dependency between the first two columns ($\mathrm{Col}_1(\mathbf{A}) = c\ \mathrm{Col}_2(\mathbf{A})$); (b) **Concurrent lines of action**: when the lines of action of all three directors intersect at a common focal point $P_c(x_c, y_c)$, the relation $t_{kx}(y_k - y_c) - t_{ky}(x_k - x_c) = 0$ renders the rotational column a linear combination of the translation columns, yielding $\mathrm{Col}_3(\mathbf{A}) = y_c\, \mathrm{Col}_1(\mathbf{A}) - x_c\, \mathrm{Col}_2(\mathbf{A})$, causing $\det(\mathbf{A}) = 0$; (c) **Shared line of action**: nodes $\mathbf{x}_1, \mathbf{x}_2$ lying on a common line $\mathcal{L}$ with $\vec{t}_1 = \vec{t}_2 \parallel (\mathbf{x}_2 - \mathbf{x}_1)$ share identical directional moments, $y_2 t_{2x} - x_2 t_{2y} = y_1 t_{1x} - x_1 t_{1y}$, producing duplicate rows $\mathrm{Row}_2(\mathbf{A}) = \mathrm{Row}_1(\mathbf{A})$; (d) **Nodal coincidence / spatial collapse**: as $\|\mathbf{x}_1 - \mathbf{x}_2\| \to 0$, if $\vec{t}_1 = \vec{t}_2$ identical rows are formed; if $\vec{t}_1 \neq \vec{t}_2$, two distinct vector degrees of freedom are colocated at a single spatial point, degenerating the 2D spatial sampling base into a 1D line; (e) **Decoupling of spatial geometry and vector alignment**: illustrates why geometric proximity algorithms alone fail—(top) spatially non-collinear nodes ($\mathrm{Area}(T) > 0$) with parallel directors yield $\det(\mathbf{A}) = 0$; (bottom) collinear nodes ($\mathrm{Area}(T) = 0$) with non-parallel directors can yield $\det(\mathbf{A}) \neq 0$ (algebraically invertible), yet fail to span 2D transverse derivatives; (f) **Admissible and stable support**: dual qualification requiring non-collinear spatial distribution ($\mathrm{Area}(T) > 0$) and non-vanishing circulation around evaluation point P, ensuring $|\det(\mathbf{A})| \geq \mathrm{Tol}_{\det}(h)$ and $\kappa(\mathbf{A}) \sim \mathcal{O}(1)$ for accurate field and curl reconstruction.

3.1.1. Decoupling of Spatial Geometry and Vector Alignment

A crucial theoretical insight governing vector meshless approximations is the formal decoupling between spatial sampling quality and vector algebraic independence, as explicitly highlighted in Fig. 2e:

- **Spatially Ideal but Algebraically Singular (Fig. 2e, top):** A candidate triad can form an optimal 2D spatial distribution (e.g., an equilateral triangle enclosing a large physical area $|T| > 0$). A purely spatial geometric filter or distance-based k-d tree query would readily accept this support. However, if the assigned vector directors are globally parallel ($\vec{t}_1 \parallel \vec{t}_2 \parallel \vec{t}_3$), matrix $\mathbf{A}$ is strictly singular ($\det(\mathbf{A}) = 0$), rendering shape function construction impossible.
- **Algebraically Solvable but Spatially Degenerate (Fig. 2e, bottom):** Conversely, if three support nodes lie on a perfectly collinear 1D line in space (zero triangle area $|T| = 0$) but possess non-parallel directors, matrix $\mathbf{A}$ in the $\mathcal{L}^1$ basis can yield a non-zero determinant ($\det(\mathbf{A}) \neq 0$). While algebraically solvable, this configuration lacks 2D spatial sampling normal to the line of nodes, leading to unconstrained polynomial extrapolation and catastrophic derivative errors in field operators.

In contrast to the pathological or decoupled configurations depicted in Figs. 2(a)–(e), an admissible and algebraically robust support—designated as an ideal support triad in Fig. 2(f)—requires a dual geometric-directional qualification. Geometrically, the candidate support nodes must form a non-degenerate 2D spatial distribution ($|T| > 0$), shielding the local domain from nodal collapse or spatial collinearity. Directionally, the assigned vector directors $\vec{t}_k$ must provide coherent tangential circulation around the evaluation point $P(x_P, y_P)$, maximizing the directional moment $M_k = (\Delta\mathbf{x}_k \times \vec{t}_k)_z \gg 0$. By enforcing non-zero spatial area coverage and vector circulation simultaneously, the scale-dependent selection algorithm rejects locally degenerate configurations and enforces the criterion $|\det(\mathbf{A})| \geq \mathrm{Tol}_{\mathrm{det}}(h)$, thereby reducing the risk of ill-conditioned local placement matrices and promoting stable shape-function construction.

3.2. Singularity Analysis in the Complete $\mathcal{P}^1$ Vector Basis (6×6 Matrix)

To eliminate derivative modal aliasing and achieve strict first-order convergence $\mathcal{O}(h^1)$ for the field curl operator $(\nabla \times \vec{E}^h)_z$, Section 2.3 introduced

the complete linear polynomial vector space $\mathcal{P}^1 = \mathcal{P}_1 \times \mathcal{P}_1$ ($M_{\text{supp}} = 6$ support nodes). Imposing the vector Kronecker delta projection property across a candidate sextet yields a local placement system governed by the 6×6 local moment placement matrix $\mathbf{A} \in \mathbb{R}^{6\times 6}$. Evaluated in local translated coordinates centered at the evaluation point $P(x_P, y_P)$ ($\Delta x_k = x_k - x_P, \Delta y_k = y_k - y_P$), matrix $\mathbf{A}$ was derived in Eq. (34) and is restated below to facilitate the singularity analysis:

$$\mathbf{A} = \begin{bmatrix} t_{1x} & t_{1y} & \Delta x_1 t_{1x} & \Delta y_1 t_{1x} & \Delta x_1 t_{1y} & \Delta y_1 t_{1y} \\ t_{2x} & t_{2y} & \Delta x_2 t_{2x} & \Delta y_2 t_{2x} & \Delta x_2 t_{2y} & \Delta y_2 t_{2y} \\ t_{3x} & t_{3y} & \Delta x_3 t_{3x} & \Delta y_3 t_{3x} & \Delta x_3 t_{3y} & \Delta y_3 t_{3y} \\ t_{4x} & t_{4y} & \Delta x_4 t_{4x} & \Delta y_4 t_{4x} & \Delta x_4 t_{4y} & \Delta y_4 t_{4y} \\ t_{5x} & t_{5y} & \Delta x_5 t_{5x} & \Delta y_5 t_{5x} & \Delta x_5 t_{5y} & \Delta y_5 t_{5y} \\ t_{6x} & t_{6y} & \Delta x_6 t_{6x} & \Delta y_6 t_{6x} & \Delta x_6 t_{6y} & \Delta y_6 t_{6y} \end{bmatrix} \tag{40}$$

By the Null-Space Orthogonality Principle, a non-zero coefficient vector $\mathbf{c} = [a_1, a_2, b_1, b_2, b_3, b_4]^T \in \mathbb{R}^6$ belongs to $\ker(\mathbf{A}_{6\times 6})$ if and only if the corresponding affine vector field $\vec{E}(\mathbf{x}) \in \mathcal{P}^1$ (with components $E_x = a_1 + b_1\Delta x + b_2\Delta y$ and $E_y = a_2 + b_3\Delta x + b_4\Delta y$) satisfies the directional orthogonality condition $\vec{E}(\mathbf{x}_k) \cdot \vec{t}_k = 0$ across all six support nodes simultaneously.

While the complete $\mathcal{P}^1$ basis inherits the classical spatial-directional pathologies of $\mathcal{L}^1$ (such as global director parallelism and action-line concurrency), expanding the local placement space to 6 degrees of freedom creates new linear dependence modes. Specifically, algebraic analysis reveals four exclusive singularity conditions governing $\mathbf{A}_{6\times 6}$:

1. **Partial Vector Parallelism (4-Node Threshold):** In contrast to $\mathcal{L}^1$ (which requires all 3 nodes to be parallel to collapse rank), $\mathbf{A}_{6\times 6}$ becomes strictly singular ($\det(\mathbf{A}) \equiv 0$, $\text{rank}(\mathbf{A}) \leq 5$) whenever **4 or more nodes** in a candidate sextet possess parallel vector directors, with $\vec{t}_1 \parallel \vec{t}_2 \parallel \vec{t}_3 \parallel \vec{t}_4$. Under coordinate rotation aligning the common direction with $\hat{x}$ ($t_{ky} = 0$), columns 2, 5, and 6 contain zeros in the first 4 rows, confining 3 vector columns to a 2-dimensional active row space and forcing linear dependence regardless of nodal spatial locations or the remaining 2 directors.
2. **Partial Spatial Collinearity (5-Node Threshold):** If **5 of the 6 support nodes** lie on a single straight line in 2D space ($\Delta y_k = m\Delta x_k$

for $k = 1, \ldots, 5$), columns 4 ($\Delta y_k t_{kx}$) and 6 ($\Delta y_k t_{ky}$) possess non-zero entries exclusively in the 6th row ($\text{Col}_4 \propto \text{Col}_6$). This direct column proportionality forces $\det(\mathbf{A}_{6\times 6}) = 0$, regardless of the vector directors assigned to all six nodes.

3. **Total Spatial Collinearity Collapse (6 Nodes):** While 1D collinear nodes in $\mathcal{L}^1$ could yield a non-zero determinant if assigned non-parallel directors (becoming "algebraically solvable but spatially degenerate"), transitioning to $\mathcal{P}^1$ eliminates this spatial vulnerability. If all 6 support nodes are collinear in 2D space ($\Delta y_k = m\Delta x_k$), two independent column dependencies arise simultaneously ($\text{Col}_4 = m\,\text{Col}_3$ and $\text{Col}_6 = m\,\text{Col}_5$), collapsing the matrix rank to $\text{rank}(\mathbf{A}) \leq 4$, which implies $\det(\mathbf{A}_{6\times 6}) \equiv 0$ unconditionally.
4. **Azimuthal Divergence Masking ($\vec{t_k} \perp \Delta\mathbf{x}_k$):** While pure concentric circulation ($\vec{t_k} \perp \Delta\mathbf{x}_k$) represented the optimal configuration for $\mathcal{L}^1$, it induces strict singularity in $\mathcal{P}^1$. Enforcing pure azimuthal alignment sets $\Delta x_k t_{kx} + \Delta y_k t_{ky} = 0$, yielding $\text{Col}_3 = -\,\text{Col}_6$ and rendering $\det(\mathbf{A}_{6\times 6}) = 0$. Physically, pure azimuthal directors cannot sample the linear isotropic divergence mode $\vec{E}_{\text{div}} = [\Delta x, \Delta y]^T \in \mathcal{P}^1$, requiring the candidate support sextet to possess angular diversity containing both tangential and radial components.

3.3. Automated Support Selection Algorithm with Scale-Dependent Determinant Floor

The mathematical classifications established in Sections 3.1 and 3.2 demonstrate that matrix singularity in the VNMM local placement system ($\mathbf{A}\boldsymbol{\beta}_i = \mathbf{L}_i$) stems from an intricate coupling between spatial node distributions ($\Delta x_k, \Delta y_k$) and unit vector director alignments (t_{kx}, t_{ky}). Attempting to construct a purely geometric selection scheme—one that relies on explicit conditional decision trees to independently test for 4-node vector parallelism, 5-node spatial collinearity, action-line focal intersection points, and azimuthal divergence masking—leads to an impractically complex, fragile, and computationally prohibitive workflow.

In contrast, evaluating the magnitude of the determinant $|\det(\mathbf{A}_{6\times 6})|$ of the local moment placement matrix serves as a single, unified algebraic filter. Because $\mathbf{A}_{6\times 6}$ incorporates both spatial coordinates and vector director projections within a single operator, enforcing $|\det(\mathbf{A}_{6\times 6})| \geq \text{Tol}_{\text{det}}(h)$ implicitly and simultaneously shields the local domain against all forms of spatial

collapse, partial vector parallelism, action-line concurrency, and divergence masking.

3.3.1. Scale-Dependent Scaling Law, Dimensional Analysis, and Absolute Tolerance Floor

In local translated coordinates centered at the evaluation point $P(x_P, y_P)$, the first two columns of $\mathbf{A} \in \mathbb{R}^{6\times 6}$ consist of dimensionless unit director components $t_{kx}, t_{ky} \sim \mathcal{O}(1)$, while columns 3–6 contain products of coordinates and directors, $\Delta x_k t_{kx}, \Delta y_k t_{kx}, \Delta x_k t_{ky}, \Delta y_k t_{ky} \sim \mathcal{O}(h)$ [m], where h denotes the local characteristic nodal spacing. Consequently, the local placement matrix is dimensional, and its determinant scales dimensionally as $[\det(\mathbf{A}_{6\times 6})] = [L]^4$:

$$\det(\mathbf{A}_{6\times 6}) \sim \mathcal{O}(h^4). \tag{41}$$

To interpret the determinant threshold independently of physical unit choices, one may consider the normalized, dimensionless local coordinates $\hat{x}_k = \Delta x_k/h$ and $\hat{y}_k = \Delta y_k/h$. The resulting dimensionless placement matrix $\widehat{\mathbf{A}}$ satisfies $\mathbf{A} = \widehat{\mathbf{A}}\ \mathrm{diag}(1, 1, h, h, h, h)$, directly establishing:

$$\det(\mathbf{A}) = h^4 \det(\widehat{\mathbf{A}}). \tag{42}$$

Thus, a scale-invariant algebraic qualification criterion is equivalent to bounding the dimensionless determinant $|\det(\widehat{\mathbf{A}})| \geq \widehat{\mathrm{Tol}} > 0$.

Under spatial refinement ($h \to 0$), enforcing a purely proportional scaling law based on a fixed reference grid (h_{ref}):

$$\mathrm{Tol}_{\mathrm{det}}^{\mathrm{pure}}(h) = \mathrm{Tol}_{\mathrm{ref}} \cdot \left(\frac{h}{h_{\mathrm{ref}}}\right)^4 \tag{43}$$

produces an excessively permissive threshold in dense point clouds (e.g., $\mathrm{Tol}_{\mathrm{det}} \approx 1.2 \times 10^{-6}$ for $h = 0.0654\,\mathrm{m}$ over $[0, \pi]^2$). As demonstrated through the parametric numerical experiments in Section 6.3.1, such unconstrained low thresholds cause the early-stopping selection mechanism to prematurely accept candidate sextets exhibiting poor angular diversity. In terms of the dimensionless matrix, this allows $|\det(\widehat{\mathbf{A}})|$ to approach zero, which elevates the condition number $\kappa_2(\mathbf{A}) > 170$ and induces local derivative degradation and stagnation in the curl approximation.

To preserve robust matrix conditioning $\kappa_2(\mathbf{A}) \sim \mathcal{O}(1)$ across all spatial discretization scales, the proposed framework enforces a scale-dependent determinant thresholding law bounded below by an absolute tolerance floor

$\mathrm{Tol}_{\mathrm{floor}}$:

$$\mathrm{Tol}_{\mathrm{det}}(h) = \max\left(\mathrm{Tol}_{\mathrm{ref}} \cdot \left(\frac{h}{h_{\mathrm{ref}}}\right)^4,\ \mathrm{Tol}_{\mathrm{floor}}\right). \tag{44}$$

By setting $\mathrm{Tol}_{\mathrm{floor}} = 1.5 \times 10^{-5}$ for the domain $\Omega = [0, \pi]^2$, the selection pipeline is forced to dynamically expand its candidate neighborhood search whenever spatial or directional alignment degrades in dense or perturbed clouds. Dimensionally, on a general domain with characteristic dimension L_{domain}, the floor scales as $\mathrm{Tol}_{\mathrm{floor}} \propto (L_{\mathrm{domain}})^4 \widehat{\mathrm{Tol}}_{\mathrm{floor}}$. As quantitatively validated in Section 6.3.2, this floor enables the identification of an Admissible Robust Support, yielding stable local configurations and restoring monotonic error reduction in the tested refinement sequences without a significant increase in computational cost.

3.3.2. The 4-Phase Selection Pipeline

To locate well-conditioned candidate sextets across unstructured point clouds without incurring exhaustive $\mathcal{O}(N^2)$ pairwise distance evaluations, the proposed selection framework combines orthogonal k-d tree spatial queries [36, 37, 38, 31] with algebraic determinant qualification. The resulting algorithm operates through a 4-phase pipeline:

1. **Phase 1: Spatial Neighborhood Query via k-d Tree:** Using an orthogonal k-d tree binary spatial search structure constructed over all domain nodes ($\mathcal{O}(N \log N)$ pre-processing complexity), the algorithm queries the $K_{\mathrm{initial}} \geq 12$ nearest physical neighbors surrounding the evaluation point $P(x_P, y_P)$ with an average query complexity of $\mathcal{O}(\log N)$ [36, 37, 38].
2. **Phase 2: Combinatorial Candidate Generation with Progressive Anchor:** Candidate support sextets are generated systematically from the queried K_{initial} neighborhood. To prioritize geometric proximity to the evaluation point P, candidate sextets are assembled by fixing an anchor node (beginning with the node closest to P) and taking 5-combinations from subsequent nearest neighbors, ordered by increasing mean Euclidean distance.
3. **Phase 3: Local Coordinate Translation and Matrix Assembly:** For each candidate sextet, spatial coordinates are translated to the local evaluation frame ($\Delta x_k = x_k - x_P, \Delta y_k = y_k - y_P$), and the local placement matrix $\mathbf{A}_{6\times 6}$ is assembled using Eq. (40).

4. **Phase 4: Algebraic Qualification with Early Stopping:** The determinant magnitude $|\det(\mathbf{A}_{6\times 6})|$ is computed. The first candidate sextet satisfying the scale-dependent criterion $|\det(\mathbf{A}_{6\times 6})| \geq \mathrm{Tol}_{\mathrm{det}}(h)$ via Eq. (44) is immediately accepted as the active support, terminating the local search loop.

3.3.3. Computational Complexity and Combinatorial Cost Analysis

A critical aspect of practical meshless implementations concerns the algorithmic complexity of support selection [11]. Because candidate sextets are formed from a local pool of K_{initial} neighbors, the theoretical maximum number of possible sextet combinations per evaluation point is bounded by the binomial coefficient:

$$N_{\mathrm{comb}}^{\max} = \binom{K_{\mathrm{initial}}}{6}. \tag{45}$$

For the baseline initial neighborhood of $K_{\mathrm{initial}} = 12$ nodes employed in this work, Eq. (45) yields $\binom{12}{6} = 924$ potential candidate sextets. If an exhaustive search were conducted across all candidate combinations, the algebraic qualification cost per evaluation point would scale as $C_{\mathrm{eval}}^{\mathrm{exhaustive}} \approx \binom{K_{\mathrm{initial}}}{6} C_{\mathrm{det}}$, where $C_{\mathrm{det}} \sim \mathcal{O}(6^3) = \mathcal{O}(1)$ denotes the fixed floating-point cost of evaluating a 6×6 determinant. Across N_{eval} evaluation points, this would represent a substantial computational overhead.

Crucially, however, the proposed selection algorithm does *not* perform an exhaustive sweep. By coupling the progressive proximity-ordered candidate generation (Phase 2) with the **early-stopping qualification criterion** (Phase 4), the search evaluates candidate sextets sequentially and terminates immediately once an admissible support satisfying $|\det(\mathbf{A}_{6\times 6})| \geq \mathrm{Tol}_{\mathrm{det}}(h)$ is identified.

As quantitatively demonstrated by our numerical experiments (Section 6.3, Table 5), the average effective neighborhood size required to locate a qualified, well-conditioned support sextet remains remarkably small: $K_{\mathrm{avg}} \approx 6.3$ for regular distributions, expanding to only $K_{\mathrm{avg}} \approx 10.3 - 11.3$ under severe 25% random spatial jitter and fully random director orientations. Consequently, the actual number of candidate sextets evaluated in practice ($\bar{N}_{\mathrm{tested}}$) is typically on the order of $\mathcal{O}(1)$ to $\mathcal{O}(10)$, which is orders of magnitude smaller than the combinatorial upper bound of 924.

Therefore, the total computational complexity of the support qualification

stage across N_{eval} evaluation points scales as:

$$T_{\text{support}} = \underbrace{\mathcal{O}(N \log N)}_{k\text{-d tree construction}} + \sum_{p=1}^{N_{\text{eval}}} \left[\underbrace{\mathcal{O}(\log N)}_{\text{spatial query}} + \underbrace{\bar{N}_{\text{tested}}^{(p)} C_{\det(6\times 6)}}_{\text{local algebraic qualification}} \right] \approx \mathcal{O}(N_{\text{eval}} \log N). \tag{46}$$

Eq. (46) establishes that while combinatorial candidate generation provides the necessary geometric-directional degrees of freedom to bypass pathological singularities in arbitrary point clouds, the early-stopping mechanism maintains quasi-linear computational scaling, preventing combinatorial bottlenecks.

4. Variational Formulation of the Resonant Cavity Problem

To evaluate the spectral accuracy and verify the complete elimination of non-physical spurious modes in the Vector Nodal Meshless Method (VNMM 2D), we formulate the electromagnetic vector eigenvalue problem in a two-dimensional resonant cavity $\Omega = [0, \pi] \times [0, \pi] \subset \mathbb{R}^2$ with perfect electric conductor (PEC) walls $\partial\Omega$ [2].

4.1. The Transverse Electric (TE_z) Vector Eigenvalue Problem

In a homogeneous, isotropic, and lossless dielectric medium ($\epsilon_r = 1.0, \mu_r = 1.0$), time-harmonic Transverse Electric modes (TE_z) are governed by the vector double-curl Helmholtz equation for the transverse electric field $\vec{E}(\mathbf{x}) = [E_x(x, y), E_y(x, y)]^T$ [3, 26]:

$$\nabla \times \left(\frac{1}{\mu_r} \nabla \times \vec{E} \right) = \lambda \, \epsilon_r \vec{E}, \quad \text{in } \Omega \tag{47}$$

$$\hat{n} \times \vec{E} = \mathbf{0}, \quad \text{on } \partial\Omega \tag{48}$$

where $\lambda = k_0^2 = \omega^2 \mu_0 \epsilon_0$ denotes the target wavenumber eigenvalue, ω is the angular resonance frequency, and $\hat{n}$ is the outward unit normal vector to boundary $\partial\Omega$. In two dimensions, the curl operator acting on a 2D vector field yields a scalar normal component $(\nabla \times \vec{E})_z = \partial E_y/\partial x - \partial E_x/\partial y$, while the vector curl of a scalar field u_z returns a vector in the transverse plane $\nabla \times u_z = [\partial u_z/\partial y, -\partial u_z/\partial x]^T$.

The exact analytical eigenvalues for the square PEC cavity are given by [2]:

$$\lambda_{m,n}^{\text{exact}} = m^2 + n^2, \quad m, n \in \{0, 1, 2, \dots\}, \quad (m + n > 0) \tag{49}$$

4.2. Ritz-Galerkin Weak Form with Div-Curl Regularization

Let the Hilbert space of square-integrable vector fields with square-integrable curl satisfying homogeneous tangential PEC boundary conditions be defined as [3, 8]:

$$\mathcal{S} \triangleq \left\{ \vec{E} \in H(\mathbf{curl}; \Omega) \;\middle|\; \hat{n} \times \vec{E} = \mathbf{0} \text{ on } \partial\Omega \right\}. \tag{50}$$

In standard Edge Finite Element Formulations (EFEM), the discrete de Rham complex property $\mathrm{Im}(\nabla) \subset \ker(\nabla\times)$ is preserved exactly at the topological level [3, 5]. Consequently, irrotational gradient fields ($\vec{E} = \nabla\phi$) yield an identically zero curl ($\nabla \times (\nabla\phi) \equiv \mathbf{0}$) and map directly into the zero eigenvalue subspace ($\lambda = 0$), preventing spectrum contamination without auxiliary constraints [3, 6].

In contrast, point-collocated meshless vector shape functions $\vec{N}_i(\mathbf{x})$ do not form an exact discrete de Rham complex at background quadrature points, resulting in non-zero numerical curl residues ($\nabla \times (\nabla\phi) \neq \mathbf{0}$). To suppress these spurious modes without introducing saddle-point system indefiniteness or auxiliary Lagrange multipliers, we adopt a **Ritz-Galerkin weak formulation with div-curl penalty regularization** [3, 8]:

$$\int_\Omega \frac{1}{\mu_r} (\nabla\times\vec{W})_z (\nabla\times\vec{E})_z \, d\Omega + s_{\mathrm{div}} \int_\Omega (\nabla\cdot\vec{W})(\nabla\cdot\vec{E}) \, d\Omega = \lambda \int_\Omega \epsilon_r \vec{W}\cdot\vec{E} \, d\Omega, \quad \forall \vec{W} \in \mathcal{S} \tag{51}$$

where $s_{\mathrm{div}} > 0$ is the non-dimensional divergence penalty parameter [3].

Upon discretizing the electric field as $\vec{E}^h(\mathbf{x}) = \sum_{j=1}^{M} c_j \vec{N}_j(\mathbf{x})$ and test functions as $\vec{W}^h(\mathbf{x}) = \sum_{i=1}^{M} d_i \vec{N}_i(\mathbf{x})$, the second bilinear term in Eq. (51) defines the **divergence penalty matrix** $\mathbf{K}_{\mathrm{div}} \in \mathbb{R}^{M\times M}$, whose entries $K_{\mathrm{div},ij}$ are explicitly defined by:

$$K_{\mathrm{div},ij} \triangleq \int_\Omega (\nabla \cdot \vec{N}_i)(\nabla \cdot \vec{N}_j) \, d\Omega \tag{52}$$

Consequently, the global stiffness matrix $\mathbf{K}$ in the discrete generalized eigenvalue problem $\mathbf{Kc} = \lambda\mathbf{Mc}$ decomposes into the sum of the primary curl-curl stiffness matrix $\mathbf{K}_{\mathrm{curl}}$ and the scaled divergence penalty matrix $s_{\mathrm{div}}\mathbf{K}_{\mathrm{div}}$:

$$\mathbf{K} = \mathbf{K}_{\mathrm{curl}} + s_{\mathrm{div}}\mathbf{K}_{\mathrm{div}} \tag{53}$$

with individual entries of the primary curl-curl stiffness matrix defined by:

$$K_{\mathrm{curl},ij} \triangleq \int_\Omega \frac{1}{\mu_r} (\nabla \times \vec{N}_i)_z (\nabla \times \vec{N}_j)_z \, d\Omega. \tag{54}$$

The physical and algebraic role of the penalty matrix $\mathbf{K}_{\text{div}}$ scaled by s_{div} is twofold [3, 5]:

1. **Preservation of Physical Modes:** Solenoidal physical TE_z modes satisfy $\nabla \cdot \vec{E} \equiv 0$ analytically. For these modes, the divergence penalty matrix vanishes ($\mathbf{K}_{\text{div}}\mathbf{c} = \mathbf{0}$), leaving physical eigenvalues $\lambda_{m,n}$ strictly unperturbed [3, 1].
2. **Penalization of Gradient Modes:** Non-physical gradient fields satisfy $\nabla \cdot (\nabla\phi) = \Delta\phi \neq 0$. The penalty term $s_{\text{div}}\mathbf{K}_{\text{div}}\mathbf{c}$ adds artificial numerical stiffness to these states, shifting their spurious eigenvalues to high frequencies far outside the operational bandwidth ($\lambda > 50$) [3, 5, 6].

As demonstrated by the systematic parametric tuning in Section 6.4, setting $s_{\text{div}} = 6.0$ provides optimal spectrum purification without inducing numerical locking.

4.3. Integration Strategy: Cell-Based vs. Point-Based (EFG-Style)

Numerical evaluation of weak form integrals in Eq. (51) requires background cell quadrature. In early VNMM implementations, support nodes and local moment placement matrices $\mathbf{A}$ were constructed once per background cell at its centroid P_c [30, 26]. However, as demonstrated by the background cell integration analysis in Section 6, expanding shape functions from P_c across large cells ($dx > h$) induces severe Taylor truncation degradation and matrix indefiniteness.

To obtain a mesh-connectivity-free field approximation, we employ a point-based Element-Free Galerkin (EFG) integration strategy over a background integration grid [14, 37]:

1. A regular background grid of $N_c \times N_c$ quadrilateral integration cells (with $N_c = \text{round}(0.6N)$) covers the physical domain Ω.
2. At each Gauss integration point $P_g(x_g, y_g)$, the algorithm queries the global k-d tree independently to select an optimal, well-conditioned support sextet $\{P_k, \vec{t}_k\}_{k=1}^{6}$ satisfying $|\det(\mathbf{A}_{6\times6})| \geq \text{Tol}_{\text{det}}(h)$.
3. Local translated coordinates are anchored directly at the evaluation point ($\Delta x = x_k - x_g = 0$, $\Delta y = y_k - y_g = 0$), ensuring that shape functions $\vec{N}_i(P_g)$ and their derivatives $\nabla \times \vec{N}_i(P_g)$, $\nabla \cdot \vec{N}_i(P_g)$ are evaluated with maximum floating-point precision without Taylor expansion drift.

Employing a 3×3 Gauss-Legendre quadrature rule (9 points per cell) ensures an adequate integration-to-DoF ratio ($N_g/N_{\text{DoF}} \geq 2.0$), preventing spatial sub-integration and mesh aliasing [39, 10].

Classification as a Meshless Formulation. In classical meshfree literature [14, 13, 10], a distinction is drawn between methods that construct approximations free of mesh connectivity (such as EFG, PIM, and the present VNMM) and methods that evaluate weak integrals without background cells (such as nodal integration or MLPG). In the present VNMM formulation:

- The field approximation and shape functions $\vec{N}_i(\mathbf{x})$ are strictly meshless and connectivity-free: no triangulation, polygonization, or element topological data structures are created or maintained between approximation nodes.
- Background quadrilateral cells serve solely as a geometric scaffolding for evaluating numerical integrals via Gauss-Legendre quadrature, exactly as in standard Element-Free Galerkin (EFG) schemes [14, 37].
- The background integration grid does not define or constrain the interpolation topology; support selection at any point is governed exclusively by k-d tree spatial proximity and algebraic determinant qualification.

4.4. Algebraic Discretization and Generalized Eigenvalue System

Approximating the electric field as $\vec{E}^h(\mathbf{x}) = \sum_{i=1}^{M_{\text{active}}} \vec{N}_i(\mathbf{x}) c_i$ and test functions as $\vec{W}^h(\mathbf{x}) = \sum_{j=1}^{M_{\text{active}}} \vec{N}_j(\mathbf{x}) d_j$, where $c_i = \vec{E}(\mathbf{x}_i) \cdot \vec{t}_i$ represent the active directional projection DoFs, yields the sparse generalized algebraic eigenvalue system [26, 31]:

$$\mathbf{K}_{\text{red}} \mathbf{c}_{\text{red}} = \lambda \mathbf{M}_{\text{red}} \mathbf{c}_{\text{red}} \tag{55}$$

The entries of the global stiffness matrix $\mathbf{K} \in \mathbb{R}^{M \times M}$ and mass matrix $\mathbf{M} \in \mathbb{R}^{M \times M}$ are assembled as [31]:

$$K_{ij} = \int_\Omega \frac{1}{\mu_r} (\nabla \times \vec{N}_i)_z (\nabla \times \vec{N}_j)_z \, d\Omega + s_{\text{div}} \int_\Omega (\nabla \cdot \vec{N}_i)(\nabla \cdot \vec{N}_j) \, d\Omega \tag{56}$$

$$M_{ij} = \int_\Omega \epsilon_r \, \vec{N}_i \cdot \vec{N}_j \, d\Omega \tag{57}$$

Homogeneous PEC boundary conditions ($\hat{n} \times \vec{E} = \mathbf{0}$) are enforced directly by setting $c_k = 0$ for all boundary nodes whose unit directors $\vec{t}_k$ align tangent to $\partial\Omega$ ($\vec{t}_k \parallel \partial\Omega$), reducing the system size to M_{red} active internal degrees of freedom [30, 31]. While the unregularized Maxwell stiffness operator is symmetric positive-semidefinite due to the static gradient nullspace, the inclusion of the divergence penalty ($s_{\text{div}} > 0$) combined with essential PEC boundary constraints renders $\mathbf{K}_{\text{red}}$ strictly positive-definite on the discrete physical subspace, while the mass matrix $\mathbf{M}_{\text{red}}$ is natively positive-definite. This ensures that the discrete generalized eigenvalue problem involves well-behaved, sparse symmetric positive-definite (SPD) reduced matrices, allowing efficient and stable computation of the physical spectrum using Krylov-subspace shift-and-invert Arnoldi methods [40, 41], as implemented via the ARPACK solver in SciPy [42].

5. Master-Slave Conforming Hybrid FEM-VNMM Coupling

To combine the localized geometric modeling flexibility and exact topological boundary conformity of the lowest-order Edge Finite Element Method (EFEM) [7, 3] with the mesh generation freedom of the Vector Nodal Meshless Method (VNMM 2D), we establish a master-slave conforming hybrid domain decomposition strategy [43, 44]. The computational domain $\Omega \subset \mathbb{R}^2$ is partitioned into two non-overlapping subdomains, $\Omega = \Omega_{\text{FEM}} \cup \Omega_{\text{VNMM}}$, joined along a common continuous interface $\Gamma_{\text{int}} = \overline{\Omega}_{\text{FEM}} \cap \overline{\Omega}_{\text{VNMM}}$.

5.1. Geometric Interface Alignment and Orientation Matching

Geometric conformity across the domain interface Γ_{int} is enforced by placing VNMM interface vector nodes P_k at the exact physical midpoints of the adjacent boundary edges $E_k \subset \Gamma_{\text{int}}$ belonging to the FEM triangular mesh Ω_{FEM}. Furthermore, the physical unit vector director $\vec{t}_k = [t_{kx}, t_{ky}]^T$ assigned to each VNMM interface node is aligned strictly parallel to the oriented unit tangent vector of the corresponding FEM edge E_k:

$$\vec{t}_k = \frac{\mathbf{x}_{k,2} - \mathbf{x}_{k,1}}{l_k} \tag{58}$$

where $\mathbf{x}_{k,1}$ and $\mathbf{x}_{k,2}$ denote the ordered end-node coordinates of FEM edge E_k, and $l_k = \|\mathbf{x}_{k,2} - \mathbf{x}_{k,1}\|$ is its physical edge length.

5.2. Nodal-Circulatory Scale Mapping and Transformation Matrix

In EFEM, the degree of freedom associated with edge E_k represents the circulation integral $e_{\text{fem},k} = \int_{E_k} \vec{E} \cdot d\boldsymbol{\ell}$ [Volts] [7, 3]. In the VNMM 2D, the degree of freedom at midpoint node P_k represents the pointwise directional field projection $e_{\text{vnmm},k} = \vec{E}(P_k) \cdot \vec{t}_k$ [Volts/meter] [30, 26].

By the Dual Equivalence Theorem proved in Section 2.2.3, the directional field projection of a first-order affine vector field along a straight line segment is invariant along its length and equals its mean circulation density. Consequently, the circulatory master DoF $e_{\text{fem},k}$ relates to the vector nodal slave DoF $e_{\text{vnmm},k}$ via the exact dimensional scaling transformation:

$$e_{\text{fem},k} = l_k \cdot e_{\text{vnmm},k} \quad \Longleftrightarrow \quad e_{\text{vnmm},k} = \frac{e_{\text{fem},k}}{l_k} \tag{59}$$

Collecting all N_γ interface degrees of freedom, we define the linear interface transformation matrix $\mathbf{T} \in \mathbb{R}^{N_\gamma \times N_\gamma}$ as a real diagonal scaling operator with entries $T_{kk} = 1/l_k$. The vector of slave interface nodal projections $\mathbf{e}_{\gamma,\text{vnmm}}$ is thus mapped directly from the master interface circulation vector $\mathbf{e}_{\gamma,\text{fem}}$ as:

$$\mathbf{e}_{\gamma,\text{vnmm}} = \mathbf{T}\mathbf{e}_{\gamma,\text{fem}} \tag{60}$$

5.3. Coupled Global System Assembly and Variational Structure

By expressing the interface nodal DoFs $\mathbf{e}_{\gamma,\text{vnmm}}$ in terms of the master interface edge DoFs $\mathbf{e}_{\gamma,\text{fem}}$ via Eq. (60), all interface meshless unknowns are eliminated prior to global assembly without introducing auxiliary Lagrange multipliers or expanding the global system dimension [44].

5.3.1. Global Unknowns Partitioning

The global master vector of unknowns $\mathbf{u}$ is partitioned into three disjoint sets:

$$\mathbf{u} = \begin{bmatrix} \mathbf{e}_{\text{fem},i} \\ \mathbf{e}_{\gamma,\text{fem}} \\ \mathbf{e}_{\text{vnmm},v} \end{bmatrix} \in \mathbb{R}^{(N_{\text{fem},i}+N_\gamma+N_{\text{vnmm},v})} \tag{61}$$

where $\mathbf{e}_{\text{fem},i} \in \mathbb{R}^{N_{\text{fem},i}}$ represents internal FEM edge circulations [V], $\mathbf{e}_{\gamma,\text{fem}} \in \mathbb{R}^{N_\gamma}$ represents master interface edge circulations [V], and $\mathbf{e}_{\text{vnmm},v} \in \mathbb{R}^{N_{\text{vnmm},v}}$ represents internal VNMM vector nodal projections [V/m].

5.3.2. Coupled Global Matrix Assembly

Applying the Ritz-Galerkin variational formulation across subdomains Ω_{fem} and Ω_{vnmm} and substituting Eq. (60) into the interface bilinear forms yields the coupled global stiffness matrix $\mathbf{K}_{\text{hybrid}}$ and mass matrix $\mathbf{M}_{\text{hybrid}}$:

$$\mathbf{K}_{\text{hybrid}} = \begin{bmatrix} \mathbf{K}_{\text{fem},ii} & \mathbf{K}_{\text{fem},i\gamma} & \mathbf{0} \\ \mathbf{K}_{\text{fem},\gamma i} & \mathbf{K}_{\text{fem},\gamma\gamma} + \mathbf{T}^T\mathbf{K}_{\text{vnmm},\gamma\gamma}\mathbf{T} & \mathbf{T}^T\mathbf{K}_{\text{vnmm},\gamma v} \\ \mathbf{0} & \mathbf{K}_{\text{vnmm},v\gamma}\mathbf{T} & \mathbf{K}_{\text{vnmm},vv} \end{bmatrix} \tag{62}$$

$$\mathbf{M}_{\text{hybrid}} = \begin{bmatrix} \mathbf{M}_{\text{fem},ii} & \mathbf{M}_{\text{fem},i\gamma} & \mathbf{0} \\ \mathbf{M}_{\text{fem},\gamma i} & \mathbf{M}_{\text{fem},\gamma\gamma} + \mathbf{T}^T\mathbf{M}_{\text{vnmm},\gamma\gamma}\mathbf{T} & \mathbf{T}^T\mathbf{M}_{\text{vnmm},\gamma v} \\ \mathbf{0} & \mathbf{M}_{\text{vnmm},v\gamma}\mathbf{T} & \mathbf{M}_{\text{vnmm},vv} \end{bmatrix} \tag{63}$$

In the coupled global system, the bilinear forms defined in Eq. (56) for the stiffness matrix entries and Eq. (57) for the mass matrix entries apply to both the Edge Finite Element Method (Ω_{fem}) and the Vector Nodal Meshless Method (Ω_{vnmm}). Crucially, however, the divergence regularization term $s_{\text{div}}(\nabla \cdot \vec{E}_i)(\nabla \cdot \vec{E}_j)$ present in Eq. (56) is evaluated exclusively within Ω_{vnmm}. Because Whitney 1-form edge elements in EFEM inherently provide curl-conformity with weakly zero divergence in their kernel, no divergence regularization is applied within Ω_{fem} ($s_{\text{div}} = 0$).

5.3.3. Symmetry, Definiteness, and Spectral Stability

The global coupled system possesses three critical algebraic and physical properties:

1. **Exact Symmetry:** Because the underlying submatrices in both subdomains are individually symmetric and $\mathbf{T}$ is a real diagonal scaling matrix ($\mathbf{T} = \mathbf{T}^T$), matrix congruence transformations such as $\mathbf{T}^T\mathbf{K}_{\text{vnmm},\gamma\gamma}\mathbf{T}$ preserve exact global symmetry for both $\mathbf{K}_{\text{hybrid}}$ and $\mathbf{M}_{\text{hybrid}}$.
2. **Definiteness and Elimination of Saddle-Point Indefiniteness:** In contrast to Lagrange multiplier (Mortar) formulations that introduce zero diagonal blocks and indefinite saddle-point blocks of the form $\begin{bmatrix} \mathbf{K} & \mathbf{C}^T \\ \mathbf{C} & \mathbf{0} \end{bmatrix}$, the direct master-slave reduction operates as an algebraic congruence transformation on the coupled system. The global mass matrix $\mathbf{M}_{\text{hybrid}}$ is strictly Symmetric Positive-Definite (SPD). The coupled

stiffness matrix $\mathbf{K}_{\text{hybrid}}$ is symmetric positive-semidefinite (SPSD) prior to boundary constraints due to the curl-curl nullspace; upon enforcing essential PEC boundary conditions and applying divergence regularization ($s_{\text{div}} > 0$ within Ω_{vnmm}), $\mathbf{K}_{\text{hybrid}}$ becomes strictly positive-definite on the discrete physical subspace. This avoids zero-mass singular blocks and ensures that all computed physical eigenvalues remain real and positive.

3. **Spectral Purity:** By preserving tangential field continuity $\vec{E}_{\text{fem}} \cdot \vec{t} = \vec{E}_{\text{vnmm}} \cdot \vec{t}$ continuously along Γ_{int} through edge-tangent collocation, the interface transformation operator suppresses non-physical spurious infinite eigenvalues ($\lambda = \infty$) that traditionally contaminate mixed saddle-point formulations.

5.4. Comparative Analysis of Interface Coupling Strategies

To contextualize the proposed master-slave direct coupling strategy within the broader domain decomposition literature [45, 46, 47, 43, 44], four distinct interface coupling paradigms were evaluated, as summarized in Table 3:

- **Direct Master-Slave Coupling (Proposed):** Extending the direct domain decomposition principles of [43, 44], geometric conformity is enforced by placing VNMM interface nodes at the midpoints of adjacent FEM boundary edges. The proposed scaling operator $T_{kk} = 1/l_k$ achieves strict tangential field continuity $\vec{E}_{\text{fem}} \cdot \vec{t} = \vec{E}_{\text{vnmm}} \cdot \vec{t}$ while avoiding saddle-point indefiniteness, preserving symmetric positive-definite properties on the regularized constrained subspace, and maintaining spectral purity with low computational overhead.

- **Nitsche's Variational Method [48, 49]:** Accommodates non-conforming interfaces by adding symmetric boundary flux integrals and a penalty parameter γ/h_{int} to the weak form [55]. While avoiding saddle-point blocks, it requires problem-dependent tuning of γ to maintain coercivity and prevent numerical locking [49].

- **Mortar Lagrange Multipliers [51, 56]:** Introduces continuous interface traction unknowns $\boldsymbol{\lambda}(s)$ representing surface magnetic currents [52]. Although accommodating non-matching grids, it expands the global system with zero diagonal blocks, yielding indefinite saddle-point systems governed by Brezzi's inf-sup condition [50] and contaminated by spurious infinite eigenvalues ($\lambda = \infty$).

Table 3: Comparative synthesis of domain interface coupling paradigms for hybrid FEM-Meshless formulations.

Coupling Strategy	**Interface Mesh**	**Matrix Structure**	**Spurious Modes**	**Complexity**
1. Direct Master-Slave (Proposed)	Conforming	Symmetric / Definite*	None	Low
2. Nitsche's Method [48, 49]	Non-conforming	Symmetric / Definite*	None	Medium
3. Lagrange Multipliers (Mortar) [50, 51, 52]	Non-conforming	Indefinite (Saddle-Point)	Present ($\lambda = \infty$)	High
4. Partition of Unity (Blending) [53, 54]	Overlapping	Symmetric / Definite*	None	Medium

*Symmetric positive-definite on the discrete physical subspace after essential boundary enforcement and div-curl regularization; unregularized curl-curl stiffness is positive-semidefinite.

- **Partition-of-Unity Blending Zones [53, 57]:** Employs a smooth ramp function $\alpha(\mathbf{x}) \in [0, 1]$ across an overlapping transition zone Ω_{trans} to bridge FEM and meshless approximants [58, 54]. While avoiding saddle-point indefiniteness, evaluating overlapping integrals increases assembly time and quadrature complexity [54].

Therefore, the direct master-slave formulation offers a highly favorable compromise between mathematical simplicity, computational efficiency, and spectral accuracy for hybrid electromagnetic analysis.

6. Numerical Verification and Validation

This section assesses the numerical performance of the proposed Vector Nodal Meshless Method (VNMM 2D). Following established guidelines in computational mechanics and electromagnetics, the numerical campaign is systematically organized into three complementary stages:

1. **Numerical Verification (Sections 6.1–6.3):** Focused on the mathematical properties of the discretized operators, including exact dual isomorphism against Whitney 1-form finite elements, restoration of asymptotic curl convergence via the complete linear vector basis $\mathcal{P}^1$, and local algebraic placement matrix conditioning under stochastic spatial-directional jitter.
2. **Physical Validation (Sections 6.4–6.5):** Focused on practical electromagnetic boundary-value problems, including Transverse Electric (TE_z) resonant cavity eigenvalue spectra under div-curl Ritz-Galerkin penalty regularization, and conforming master-slave hybrid EFEM-VNMM domain decomposition.
3. **Limitations and Scope of Applicability (Section 6.6):** Focused on establishing honest methodological boundaries, discussing background quadrature dependencies, extreme point-cloud anisotropy, penalty parameter sensitivity, and high-frequency dispersion.

6.1. Conforming Dual Equivalence: VNMM $\mathcal{L}^1$ vs. Whitney EFEM

To establish a rigorous baseline connecting the proposed vector meshless formulation to established finite element theory, this first experiment evaluates the local vector interpolation capacity of the incomplete 3-node basis $\mathcal{L}^1$ against lowest-order Whitney 1-form edge elements (Nédélec 1st-kind) [7, 3].

6.1.1. Experimental Setup and Dual Topologies

The benchmark test considers the TE_{11} resonant mode over the square cavity domain $\Omega = [0, \pi] \times [0, \pi]$, governed by the exact analytical field [2, 1]:

$$\vec{E}_{\text{ex}}(x, y) = \begin{bmatrix} \cos(x)\sin(y) \\ -\sin(x)\cos(y) \end{bmatrix}, \qquad (\nabla \times \vec{E}_{\text{ex}})_z(x, y) = -2\cos(x)\cos(y) \quad (64)$$

To compare both formulations under identical spatial sampling, a sequence of seven conforming triangular meshes (4×4 to 32×32) is generated with controlled stochastic vertex perturbation (10% jitter) to ensure arbitrary, non-equilateral simplices. All interpolation errors are evaluated across a fixed, invariant grid of 1,600 internal Cartesian points (40×40).

The geometric arrangement of the conforming dual topology is illustrated in Fig. 3. For a triangular cell T with vertices V_1, V_2, V_3, directed edges $e_k = (V_i, V_j)$ of length L_k, and midpoints $M_k = \frac{1}{2}(V_i + V_j)$, the discrete degrees of freedom are assigned as follows:

1. **EFEM (Whitney 1-Forms):** Classical edge elements define circulatory DoFs along the directed mesh edges as $e_{\text{fem},k} = L_k(\vec{E}(M_k) \cdot \vec{t}_k)$ [Volts].
2. **VNMM 2D ($\mathcal{L}^1$):** The meshless vector nodal distribution places vector nodes precisely at the edge midpoints M_k, assigning unit directional vectors $\vec{t}_k$ aligned along the triangle edges, yielding pointwise DoFs $e_{\text{vnmm},k} = \vec{E}(M_k) \cdot \vec{t}_k$ [Volts/meter], as depicted in Fig. 3.

Following the discrete scaling formulation introduced in Section 5.2, the edge length L_k maps the circulatory DoFs of EFEM to the directional field projections of VNMM 2D ($e_{\text{fem},k} = L_k e_{\text{vnmm},k}$).

6.1.2. Numerical Identity and Asymptotic Convergence Rates

The asymptotic convergence behavior comparing the VNMM 2D $\mathcal{L}^1$ and Whitney EFEM formulations across the seven conforming mesh levels is presented in Fig. 4.

As demonstrated in Fig. 4a, the electric field RMS errors $\|\vec{E} - \vec{E}^h\|_{\text{RMS}}$ for both formulations display an identical first-order asymptotic decay rate of $\mathcal{O}(h^{1.00})$. The point-by-point field difference between the two methods remains strictly bounded by double-precision floating-point machine precision, satisfying $\|\vec{E}^{\text{VNMM}} - \vec{E}^{\text{EFEM}}\|_{\text{RMS}} \leq 1.6 \times 10^{-16}$.

Similarly, Fig. 4b illustrates the curl operator RMS error $\|(\nabla \times \vec{E})_z - (\nabla \times \vec{E}^h)_z\|_{\text{RMS}}$, which exhibits an identical fitted convergence rate of $\mathcal{O}(h^{0.96})$

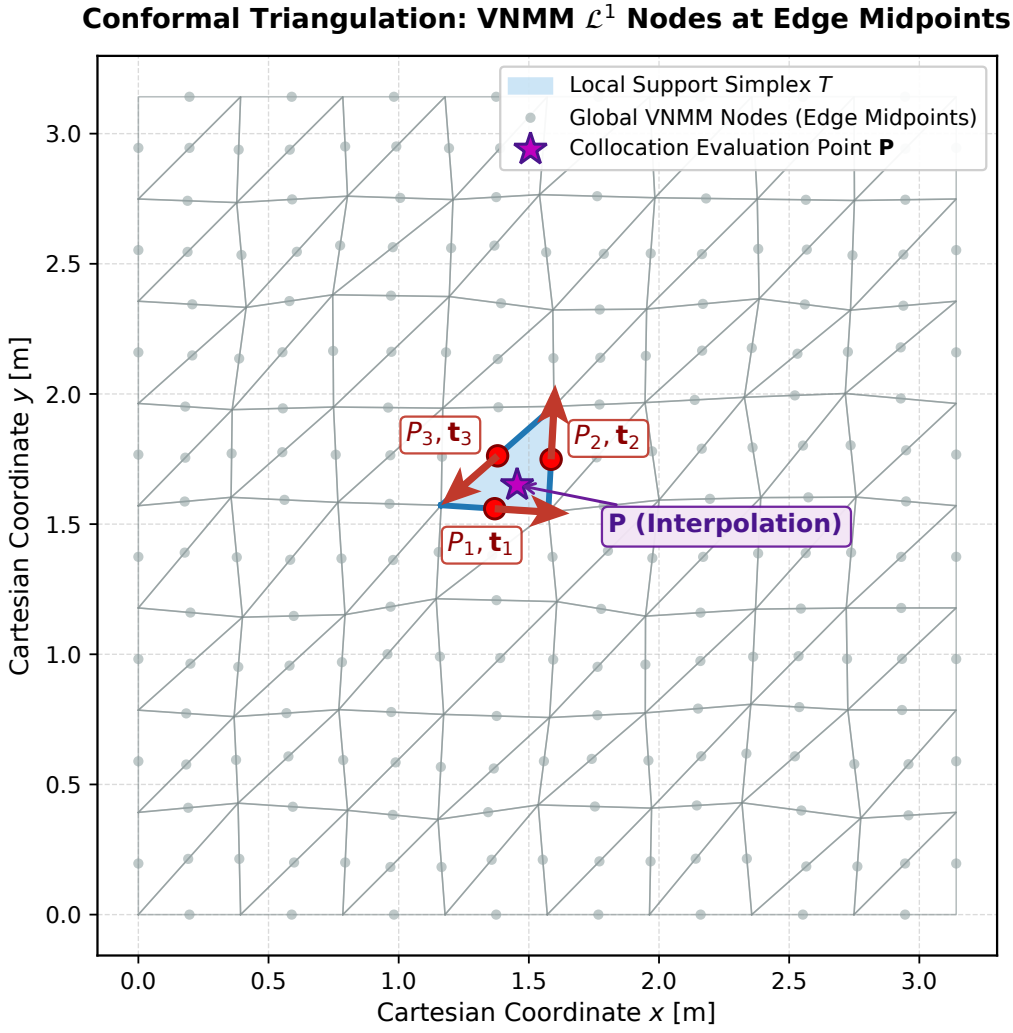


Figure 3: Conforming dual topology representation over the square domain $\Omega = [0, \pi] \times [0, \pi]$, displaying the underlying triangular finite element mesh alongside the dual VNMM $\mathcal{L}^1$ vector nodal cloud. Vector nodes are co-located at edge midpoints M_k with unit directional vectors $\vec{t}_k$ aligned along the triangle edges. This exact midpoint co-location guarantees that support triads $\{P_1, P_2, P_3\}$ form local triangles matching the EFEM cells, establishing pointwise dual equivalence.

for both formulations. The mutual curl field difference is likewise bounded by machine precision, satisfying $\|(\nabla\times\vec{E})^{\mathrm{VNMM}}-(\nabla\times\vec{E})^{\mathrm{EFEM}}\|_{\mathrm{RMS}}\leq 2.5\times10^{-15}$. These log-log convergence curves empirically confirm that the VNMM 2D $\mathcal{L}^1$ acts as the exact pointwise dual operator of Whitney 1-forms on conforming triangular cells.

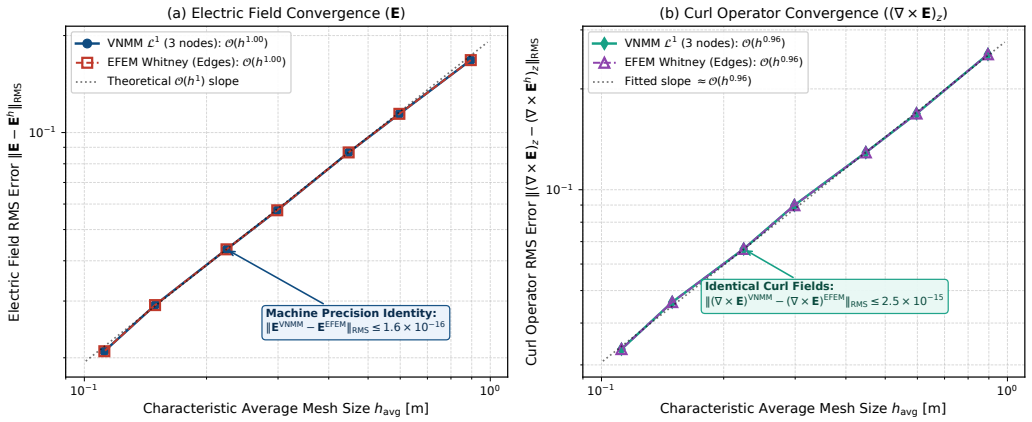


Figure 4: Asymptotic convergence rates comparing the Edge Finite Element Method (EFEM, Whitney 1-forms) and the Vector Nodal Meshless Method (VNMM $\mathcal{L}^1$, 3 support nodes) evaluated over the exact same conforming triangular mesh sequence: (a) electric field RMS error $\|\vec{E}-\vec{E}^h\|_{\mathrm{RMS}}$ exhibiting strictly identical $\mathcal{O}(h^{1.00})$ convergence with machine-precision difference $\|\vec{E}^{\mathrm{VNMM}}-\vec{E}^{\mathrm{EFEM}}\|_{\mathrm{RMS}}\leq 1.6\times10^{-16}$; (b) curl operator RMS error $\|(\nabla\times\vec{E})_z-(\nabla\times\vec{E}^h)_z\|_{\mathrm{RMS}}$ displaying identical $\mathcal{O}(h^{0.96})$ convergence with mutual difference $\|(\nabla\times\vec{E})^{\mathrm{VNMM}}-(\nabla\times\vec{E})^{\mathrm{EFEM}}\|_{\mathrm{RMS}}\leq 2.5\times10^{-15}$, confirming the exact constructive isomorphism between Whitney edge elements and VNMM $\mathcal{L}^1$.

Crucially, this exact mathematical equivalence holds **strictly under conforming mesh topologies** where support triads $\{P_1,\ P_2,\ P_3\}$ are constrained to edge midpoints of a well-defined cell. When transitioning to unconstrained, arbitrary meshless point clouds, this equivalence breaks down, as evaluated next.

6.2. Vector Field Interpolation and Derivative Convergence in Arbitrary Meshless Clouds

While Section 6.1 established the exact pointwise equivalence between the incomplete 3-node basis $\mathcal{L}^1$ and Whitney 1-form edge elements on conforming triangular meshes, practical meshless modeling requires evaluating fields over unconstrained, non-conforming point clouds where vector nodes do not

represent edge midpoints of an underlying triangulation [32, 31]. This subsection evaluates the asymptotic convergence behavior of the VNMM 2D under arbitrary spatial node distributions, validating the modal aliasing analysis presented in Section 2.2.5 and demonstrating its resolution via the 6-node linear polynomial vector space $\mathcal{P}^1$.

6.2.1. Experimental Setup on Unconstrained Point Clouds

The interpolation performance is assessed for the analytical TE_{11} cavity mode specified in Eq. (64) over unstructured, non-conforming nodal distributions covering $\Omega = [0, \pi] \times [0, \pi]$. Boundary nodes are placed uniformly along the domain perimeter with counterclockwise unit tangent vectors, whereas internal nodes are generated via a stratified Cartesian grid with 25% stochastic spatial perturbation (*jitter*) and random directional vector orientations $\vec{t}_k = [\cos\theta_k, \sin\theta_k]^T$.

To ensure strict statistical consistency, all field and derivative errors are evaluated across a fixed, invariant Cartesian grid of 1,600 internal sampling points (40×40) distributed uniformly in $[0.04\pi, 0.96\pi]^2$. The computational domain is discretized over six refinement levels, with node counts $N \in \{84, 186, 416, 884, 1928, 4192\}$ and average nodal spacings ranging from $h_{\text{avg}} = 0.5236$ m down to 0.0654 m ($h_{\text{avg}} \in \{0.52, 0.35, 0.22, 0.15, 0.10, 0.07\}$ m).

6.2.2. Empirical Validation of Curl Stagnation in $\mathcal{L}^1$

When evaluated on unconstrained meshless clouds, the 3-node solenoidal basis $\mathcal{L}^1$ exhibits the fundamental breakdown in derivative convergence predicted theoretically in Section 2.2.5. As illustrated by the log-log curves in Fig. 5a, the electric field RMS error $\|\vec{E} - \vec{E}^h\|_{\text{RMS}}$ under $\mathcal{L}^1$ decreases monotonically at a first-order rate of $\mathcal{O}(h^{0.97})$, dropping from 1.91×10^{-1} ($h = 0.5236$ m) down to 2.46×10^{-2} ($h = 0.0654$ m).

However, as revealed in Fig. 5b, the RMS error of the curl operator $\|(\nabla \times \vec{E})_z - (\nabla \times \vec{E}^h)_z\|_{\text{RMS}}$ **stagnates at an $\mathcal{O}(1)$ error plateau** (≈ 0.85, varying between 8.33×10^{-1} and 8.88×10^{-1}, asymptotic rate $\mathcal{O}(h^{-0.01}) \approx \mathcal{O}(1)$), failing to converge under spatial refinement ($N = 4192$). This persistent noise floor confirms the modal aliasing mechanism derived in Section 2.2.5, wherein unrepresented normal field derivatives leak into the rotational coefficient via the inverted placement matrix $\mathbf{A}^{-1}$.

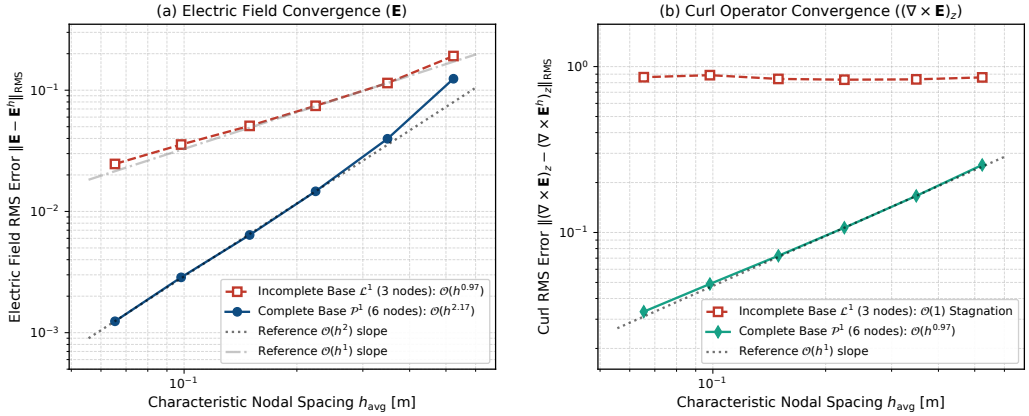


Figure 5: Log-log convergence comparison between the incomplete Whitney-like base $\mathcal{L}^1$ (3 support nodes) and the complete linear base $\mathcal{P}^1$ (6 support nodes) over $\Omega = [0, \pi] \times [0, \pi]$ evaluated across a fixed Cartesian grid of 1,600 internal points for the analytical TE_{11} cavity mode: (a) electric field RMS error displaying standard $\mathcal{O}(h^{0.97})$ convergence for $\mathcal{L}^1$ versus an observed rate of $\mathcal{O}(h^{2.17})$ for $\mathcal{P}^1$, delivering a 19.8× error reduction on the finest grid ($h = 0.0654$ m); (b) curl operator RMS error demonstrating $\mathcal{O}(1)$ error stagnation for $\mathcal{L}^1$ (plateau at ≈ 0.85 due to missing cross-derivatives), in contrast to asymptotic $\mathcal{O}(h^{0.97})$ convergence for $\mathcal{P}^1$, yielding a 25.9× error reduction.

6.2.3. Restoration of Asymptotic Convergence via Complete $\mathcal{P}^1$ Basis

Expanding the local approximation space to the complete linear polynomial vector space $\mathcal{P}^1$ introduced in Section 2.3 provides six independent degrees of freedom, capturing all four components of the local Jacobian tensor $\nabla \vec{E}^h$. Local support domains (averaging $K_{\text{avg}} \approx 11.0$ neighbors, ranging from 10.7 to 11.2) are selected via k-d tree spatial searches subject to the adaptive quartic determinant threshold developed in Section 3.3.

As demonstrated in Fig. 5, transitioning to $\mathcal{P}^1$ removes derivative modal aliasing and achieves two major performance gains:

1. **Observed Quadratic Convergence for Field $\vec{E}^h$ (Fig. 5a):** Completing the linear polynomial space elevates the observed electric field convergence rate from $\mathcal{O}(h^{0.97})$ to approximately quadratic $\mathcal{O}(h^{2.17})$. At the finest discretization ($h = 0.0654$ m), the field RMS error drops from 2.46×10^{-2} ($\mathcal{L}^1$) down to 1.24×10^{-3} ($\mathcal{P}^1$), representing a 19.8× **precision gain**.
2. **Restoration of Asymptotic $\mathcal{O}(h^1)$ Curl Convergence (Fig. 5b):** By accommodating normal derivatives in the Taylor expansion, the

complete $\mathcal{P}^1$ basis restores first-order convergence $\mathcal{O}(h^{0.97})$ for the curl operator. The curl RMS error decreases monotonically from 2.55×10^{-1} down to 3.33×10^{-2} at $h = 0.0654$ m, delivering a 25.9× **error reduction** over the stagnated $\mathcal{L}^1$ formulation (8.63×10^{-1}).

Consequently, while $\mathcal{L}^1$ is suitable for conforming triangular meshes where circulation cancellation holds (Section 6.1), the complete $\mathcal{P}^1$ space is mathematically indispensable for unconstrained, arbitrary meshless point clouds. The quantitative performance comparison between the two bases across all six refinement levels is detailed in Table 4.

Table 4: Comparative vector interpolation error metrics and asymptotic convergence between the incomplete base $\mathcal{L}^1$ (3 support nodes) and complete base $\mathcal{P}^1$ (6 support nodes) over $\Omega = [0, \pi] \times [0, \pi]$ evaluated across a fixed grid of 1,600 internal points for the analytical TE_{11} cavity mode.

N_{total}	h_{avg} [m]	**Electric Field RMS Error**			**Curl Operator RMS Error**			K_{avg}
		$\mathcal{L}^1$	$\mathcal{P}^1$	**Gain**	$\mathcal{L}^1$	$\mathcal{P}^1$	**Gain**	($\mathcal{P}^1$)
84	0.5236	1.91×10^{-1}	1.24×10^{-1}	**1.5**×	8.61×10^{-1}	2.55×10^{-1}	**3.4**×	11.1
186	0.3491	1.15×10^{-1}	3.97×10^{-2}	**2.9**×	8.38×10^{-1}	1.66×10^{-1}	**5.0**×	11.2
416	0.2244	7.44×10^{-2}	1.47×10^{-2}	**5.1**×	8.34×10^{-1}	1.07×10^{-1}	**7.8**×	11.2
884	0.1496	5.08×10^{-2}	6.39×10^{-3}	**7.9**×	8.43×10^{-1}	7.24×10^{-2}	**11.6**×	10.9
1928	0.0982	3.56×10^{-2}	2.86×10^{-3}	**12.5**×	8.88×10^{-1}	4.89×10^{-2}	**18.1**×	10.9
4192	0.0654	2.46×10^{-2}	1.24×10^{-3}	**19.8**×	8.63×10^{-1}	3.33×10^{-2}	**25.9**×	10.7
Fitted Rate		$\mathcal{O}(h^{0.97})$	$\mathcal{O}(\mathbf{h^{2.17}})$	—	$\mathcal{O}(h^{-0.01})$	$\mathcal{O}(\mathbf{h^{0.97}})$	—	—

6.3. Numerical Robustness Under Stochastic Spatial-Directional Perturbations

While Section 6.2 established the mathematical necessity of the complete $\mathcal{P}^1$ space to eliminate derivative modal aliasing, practical meshless computing on arbitrary unstructured point clouds demands evaluating the algebraic stability of the support qualification algorithm itself [11]. In unconstrained clouds, arbitrary spatial arrangements and director orientations can induce pathological local configurations—such as near-parallel directors (≥ 4 parallel nodes) or near-collinear nodes (≥ 5 collinear nodes)—that degrade the conditioning of the local placement matrix $\mathbf{A} \in \mathbb{R}^{6\times6}$ (Section 3.2).

To evaluate the stabilization mechanism provided by the scale-dependent determinant thresholding law with an absolute floor (Eq. (44), Section 3.3) within the complete $\mathcal{P}^1$ basis over $\Omega = [0, \pi] \times [0, \pi]$, we subject the point

cloud sequence to a representative benchmark scenario combining simultaneous severe stochastic spatial-directional distortions:

1. **Stochastic Spatial Jitter:** Interior node coordinates are perturbed as $\mathbf{x}_k \leftarrow \mathbf{x}_k + \delta\mathbf{x}_k$, with independent random displacements δx_k, $\delta y_k \sim \mathcal{U}(-0.25\,\Delta x, 0.25\,\Delta x)$, introducing a 25% random spatial jitter relative to the base lattice spacing.
2. **Random Unit Director Orientations:** Unit vector directors assigned to interior nodes receive fully random angles $\theta_k \sim \mathcal{U}(0, 2\pi)$, defining $\vec{t_k} = [\cos\theta_k, \sin\theta_k]^T$, completely removing directional alignment symmetry.

This benchmark realization serves as a stringent test bed for the automated qualification pipeline, assessing whether the scale-dependent determinant threshold can dynamically detect and bypass localized pathological candidate sextets without prior manual intervention. Figure 6 illustrates the resulting convergence behavior and parametric sensitivity analysis comparing the unconstrained pure quartic scaling law against the stabilized formulation with an absolute determinant floor.

6.3.1. Failure of the Pure Quartic Scaling Law in Dense Perturbed Clouds (Fig. 6a)

As established in Section 3.2, placement matrix columns 3–6 in the complete $\mathcal{P}^1$ basis scale linearly with local node spacing as $\mathcal{O}(h)$, causing the local placement determinant to scale quartically as $\det(\mathbf{A}_{6\times6}) \sim \mathcal{O}(h^4)$. When employing a pure unconstrained scaling law $\mathrm{Tol}_{\mathrm{det}}^{\mathrm{pure}}(h) \propto h^4$, the determinant acceptance threshold drops precipitously in dense grids, reaching $\mathrm{Tol}_{\mathrm{det}} \approx 1.2 \times 10^{-6}$ at $h = 0.0654\,\mathrm{m}$ ($N = 4{,}192$).

Under stochastic spatial-directional noise, this unconstrained drop causes numerical instability:

- At $\mathrm{Tol}_{\mathrm{det}} \leq 1.2 \times 10^{-6}$, the early-stopping qualification accepts the immediate 6 nearest spatial neighbors ($K_{\mathrm{avg}} \approx 6.3$).
- Because node coordinates and directors are stochastically perturbed, these immediate spatial neighbors frequently form ill-conditioned sextets approaching the singularity modes analyzed in Section 3.2 (e.g., 4 near-parallel directors or 5 near-collinear nodes), elevating the matrix condition number to $\kappa(\mathbf{A}) > 180$.

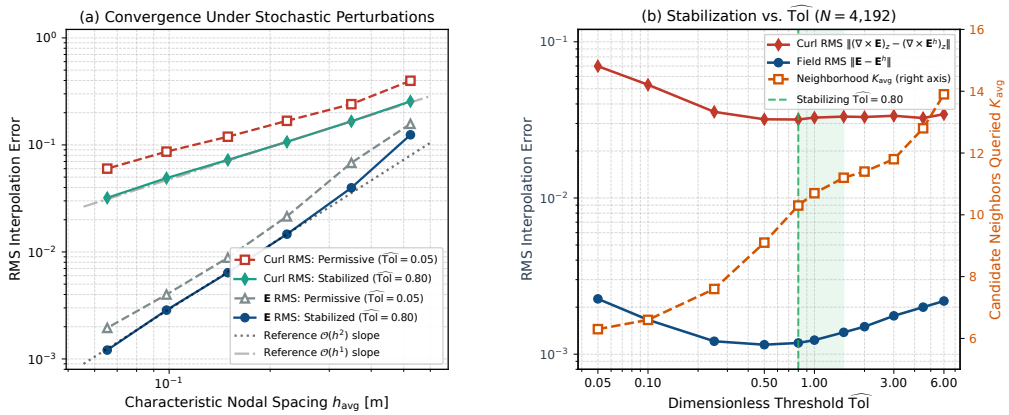


Figure 6: Numerical robustness and conditioning analysis for the 2D VNMM complete base $\mathcal{P}^1$ under stochastic perturbations (uniform nodal jitter of 25% and random director angles $\theta_i \in [0, 2\pi)$) over $\Omega = [0, \pi]^2$: (a) Convergence under stochastic mesh refinement with and without the determinant tolerance floor, highlighting curl degradation when scaling $\mathrm{Tol}_{\mathrm{det}} \propto h^4$ without a floor due to ill-conditioned sextets ($\kappa(\mathbf{A}) > 180$) and restored strict monotonic convergence when applying $\mathrm{Tol}_{\mathrm{floor}} = 1.5 \times 10^{-5}$; (b) Interpolation error and candidate search neighborhood K_{avg} as a function of the threshold criterion $\mathrm{Tol}_{\mathrm{det}}$ for the fine grid ($N = 4{,}192$), identifying the stable operating regime for $\mathrm{Tol}_{\mathrm{det}} \geq 1.5 \times 10^{-5}$.

- Because spatial derivatives scale dimensionally as $\mathcal{O}(h^{-1})$, the ill-conditioned inverted matrix $\mathbf{A}^{-1}$ amplifies local interpolation errors. As shown by the dashed crimson curve in Fig. 6a, this causes the curl operator RMS error to flatten and stagnate as $h \to 0.0654$ m, remaining elevated at an RMS error of $\approx 6.01 \times 10^{-2}$ to 8.66×10^{-2} for $(\nabla \times \vec{E}^h)_z$.

6.3.2. Restoration of Monotonicity via Absolute Tolerance Floor (Fig. 6a)

Enforcing the absolute lower bound $\mathrm{Tol}_{\mathrm{floor}} = 1.5 \times 10^{-5}$ in the scale-dependent law (Eq. (44), Section 3.3) mitigates this derivative instability.

As demonstrated by the solid teal curves in Fig. 6a, when $h \to 0.0654\,\mathrm{m}$, the lower bound $\mathrm{Tol}_{\mathrm{floor}} = 1.5 \times 10^{-5}$ overrides the decaying quartic threshold. This forces the k-d tree candidate search to expand its local sampling radius moderately (K_{avg} expands from 6.3 to 10.3), automatically skipping near-collinear or near-parallel neighbor combinations that violate the determinant floor. By dynamically identifying an Admissible Robust Support via the 4-phase selection algorithm (Section 3.3.2), the method restores monotonic error reduction across the tested refinement range for both field $\vec{E}^h$ and curl $(\nabla \times \vec{E}^h)_z$, reducing curl error to 3.20×10^{-2} on the finest mesh ($N = 4{,}192$).

6.3.3. Parametric Neighborhood Trade-off and Stabilization Threshold (Fig. 6b)

To determine the optimal stabilization threshold on $\Omega = [0, \pi]^2$, a systematic parametric sweep was conducted on the stochastically perturbed fine grid ($N = 4{,}192, h = 0.0654\,\mathrm{m}$) across determinant thresholds ranging from $\mathrm{Tol}_{\mathrm{det}} = 1.0 \times 10^{-6}$ up to 1.0×10^{-4}. The quantitative results are summarized in Table 5.

Table 5: Parametric error sensitivity, support neighborhood size K_{avg}, and placement matrix conditioning as a function of determinant threshold $\mathrm{Tol}_{\mathrm{det}}$ for the stochastically perturbed fine grid ($N = 4{,}192, h = 0.0654\,\mathrm{m}$) over $\Omega = [0, \pi]^2$.

Threshold	**Success**	**Neighbors**	**RMS Error**		**Conditioning**
$\mathrm{Tol}_{\mathrm{det}}$	**Rate**	K_{avg}	$\vec{E}^h$	$(\nabla \times \vec{E}^h)_z$	$\kappa_2(\mathbf{A})$
1.0×10^{-6} (Permissive)	100%	6.3	2.21×10^{-3}	6.86×10^{-2}	> 180 (Sub-cond.)
3.0×10^{-6}	100%	7.0	1.34×10^{-3}	4.09×10^{-2}	≈ 141
8.0×10^{-6}	100%	8.8	1.15×10^{-3}	3.27×10^{-2}	≈ 99
$\mathbf{1.5 \times 10^{-5}}$ **(Floor)**	100%	**10.3**	$\mathbf{1.21 \times 10^{-3}}$	$\mathbf{3.20 \times 10^{-2}}$	$\approx$ **73** **(Stable)**
$\mathbf{3.0 \times 10^{-5}}$ **(Robust)**	100%	**11.3**	$\mathbf{1.41 \times 10^{-3}}$	$\mathbf{3.33 \times 10^{-2}}$	$\approx$ **64** **(Robust)**
6.0×10^{-5}	100%	11.9	1.80×10^{-3}	3.33×10^{-2}	≈ 61
1.0×10^{-4}	100%	13.5	2.09×10^{-3}	3.46×10^{-2}	≈ 58

As illustrated in Fig. 6b:

1. **Stabilization Zone ($\mathrm{Tol}_{\mathrm{floor}} \geq 1.5 \times 10^{-5}$):** Increasing $\mathrm{Tol}_{\mathrm{det}}$ from 1.0×10^{-6} to 1.5×10^{-5} produces a sharp, monotonic reduction in curl interpolation error (a $2.1\times$ precision gain from 6.86×10^{-2} down to 3.20×10^{-2}), while maintaining bounded matrix conditioning ($\kappa_2(\mathbf{A}) \leq 73$, reaching ≈ 64 at $\mathrm{Tol}_{\mathrm{det}} = 3.0 \times 10^{-5}$).
2. **Compact Support Locality Invariance ($K_{\mathrm{avg}} \leq 11.3$):** This precision gain requires expanding the candidate search neighborhood from $K_{\mathrm{avg}} = 6.3$ to only $K_{\mathrm{avg}} = 10.3 - 11.3$ nodes. Crucially, this value matches the average neighborhood size observed on unperturbed clouds ($K_{\mathrm{avg}} \approx 11.0$, Table 4), demonstrating that the absolute tolerance floor effectively suppresses directional collinearities without unphysical support dilation, thereby preserving spatial search efficiency and compact support locality without combinatorial overhead (Section 3.3.3).

While this benchmark demonstrates the mechanical action of the determinant floor across an individual perturbed refinement sequence, a full Monte Carlo characterization involving ensembles of tens of stochastic realizations per refinement level constitutes an important avenue for future statistical validation of generalized unstructured cloud generators.

6.4. Resonant Cavity Spectral Analysis and Div-Curl Regularization

Having established the variational framework for the electromagnetic vector eigenvalue problem in Section 4, this experiment evaluates the spectral accuracy of the Vector Nodal Meshless Method (VNMM 2D) and verifies the elimination of non-physical spurious modes. Numerical simulations are conducted for Transverse Electric (TE_z) modes over the square PEC cavity using the complete $\mathcal{P}^1$ basis, comparing discrete spectra against the analytical reference eigenvalues defined in Eq. (49).

Figure 7 presents the discrete eigenvalue spectra ($\lambda = k_c^2$), contrasting the unregularized formulation against the div-curl regularized formulation ($s_{\text{div}} = 6.0$) developed in Eq. (51).

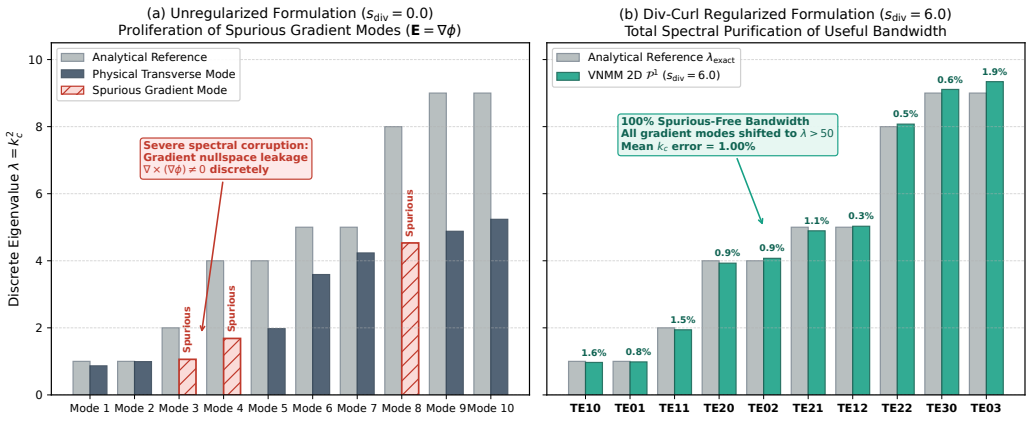


Figure 7: Discrete eigenvalue spectrum $\lambda = k_c^2$ for the TE_z square PEC cavity: (a) Unregularized curl-curl formulation ($s_{\text{div}} = 0.0$), illustrating the severe spectral contamination caused by discrete gradient nullspace leakage ($\nabla \times (\nabla\phi) \neq \mathbf{0}$), where non-zero spurious modes infiltrate the physical sequence; (b) Div-curl regularized formulation ($s_{\text{div}} = 6.0$), demonstrating effective spectral purification across the useful physical bandwidth with no spurious modes observed among the tested resonances, shifting gradient modes to $\lambda > 50$ and restricting the mean cutoff wavenumber error to 1.00%.

Numerical evaluations are conducted on the base-case regular discretization with $N_x = 21, N_y = 21$ ($N_{\text{total}} = 441$ nodes, yielding $M_{\text{active}} = 361$

internal degrees of freedom after enforcing homogeneous tangential PEC boundary conditions $c_k = 0$ on $\partial\Omega$). The weak form integrals are computed over a background grid of 10×10 quadrilateral cells ($N_c = 100$) using 2×2 Gauss-Legendre quadrature (400 Gauss points), maintaining a suitable quadrature-to-DoF ratio ($N_g/N_{\text{DoF}} \approx 1.11$) [14, 12, 10].

6.4.1. Spectral Contamination in the Unregularized Curl-Curl Formulation ($s_{\text{div}} = 0.0$)

As analyzed theoretically in Section 4.2, point-collocated meshless vector shape functions fail to form an exact discrete de Rham complex at background quadrature points. Unlike Edge Finite Element Methods (EFEM) that achieve closed-contour circulation cancellation, discrete gradient fields in unregularized meshless formulations generate non-zero numerical curl residues ($\nabla \times (\nabla\phi) \neq \mathbf{0}$ at Gauss points).

The empirical impact of this nullspace leakage is clearly observed in Fig. 7a. Setting $s_{\text{div}} = 0.0$ transforms analytical zero-eigenvalues belonging to $\ker(\nabla\times)$ into small, positive non-zero states ($0.8 < \lambda < 5.0$). These non-physical spurious gradient modes infiltrate the physical TE_z sequence, corrupting modal ordering and elevating the mean relative cutoff wavenumber error to unacceptable levels (19.75% to 90.0%).

6.4.2. Spectral Purification via Div-Curl Penalty Regularization ($s_{\text{div}} = 6.0$)

To eliminate the non-physical spurious gradient modes identified in Section 6.4.1 without introducing auxiliary Lagrange multipliers or expanding the system matrix, the div-curl variational penalty parameter is activated with $s_{\text{div}} = 6.0$ in Eq. (51).

As established theoretically in Section 4.2, the divergence penalty matrix $\mathbf{K}_{\text{div}}$ operates selectively based on solenoidal field identity: physical TE_z cavity modes satisfy $\nabla \cdot \vec{E} \equiv 0$ analytically, leaving their physical eigenvalues strictly unperturbed, whereas irrotational gradient states ($\nabla \cdot (\nabla\phi) = \Delta\phi \neq 0$) receive artificial numerical stiffness that shifts their spurious eigenvalues far outside the operational frequency window ($\lambda > 50$).

The experimental confirmation of this selective purification is demonstrated in Fig. 7b. Activating $s_{\text{div}} = 6.0$ yields spurious-free modal pairing across the operational bandwidth, removing spurious branches from the low-frequency discrete spectrum and restoring 1-to-1 modal correspondence with the analytical reference eigenvalues defined in Eq. (49) for all investigated modes.

6.4.3. Modewise Validation and Cutoff Wavenumber Precision

The numerical performance of the regularized VNMM 2D ($\mathcal{P}^1$, $s_{\text{div}} = 6.0$) for the first 10 physical TE_{nm} cavity modes is detailed in Table 6.

Table 6: Comparison of the first 10 physical TE_{nm} resonant modes in the square PEC cavity $\Omega = [0, \pi]^2$ between analytical reference values and the proposed regularized VNMM 2D formulation ($\mathcal{P}^1$, $s_{\text{div}} = 6.0$, 361 active DoFs).

Mode (TE_{nm})	$\lambda_{\text{analytical}}$	$k_{c,\text{analytical}}$	λ_{VNMM}	$k_{c,\text{VNMM}}$	**Error** λ (%)	**Error** k_c (%)
TE_{10}	1.00	1.000	0.9679	0.984	3.21%	**1.62**%
TE_{01}	1.00	1.000	0.9836	0.992	1.64%	**0.82**%
TE_{11}	2.00	1.414	1.9409	1.393	2.95%	**1.49**%
TE_{20}	4.00	2.000	3.9275	1.982	1.81%	**0.91**%
TE_{02}	4.00	2.000	4.0724	2.018	1.81%	**0.90**%
TE_{21}	5.00	2.236	4.8925	2.212	2.15%	**1.08**%
TE_{12}	5.00	2.236	5.0287	2.242	0.57%	**0.29**%
TE_{22}	8.00	2.828	8.0743	2.842	0.93%	**0.46**%
TE_{30}	9.00	3.000	9.1089	3.018	1.21%	**0.60**%
TE_{03}	9.00	3.000	9.3409	3.056	3.79%	**1.88**%
Global Mean	—	—	—	—	**2.01**%	**1.00**%

The regularized VNMM 2D achieves a global mean cutoff wavenumber error of 1.00% (with a maximum error of 1.88% for the TE_{03} mode), strictly matching analytical degeneracies ($TE_{10}/TE_{01}, TE_{20}/TE_{02}, TE_{21}/TE_{12}, TE_{30}/TE_{03}$) without spurious mode infiltration. Importantly, parametric sensitivity evaluations confirm that this spectral purification is highly robust across a wide penalty window $s_{\text{div}} \in [4.0, 10.0]$, verifying that the formulation does not require delicate, problem-dependent parameter tuning.

6.5. Comparative Convergence: Pure VNMM 2D vs. Master-Slave Hybrid vs. Edge FEM

This section presents a comparative benchmark across three distinct numerical formulations for Transverse Electric (TE_z) cavity resonance over $\Omega = [0, \pi]^2$:

1. **Pure Edge FEM (Whitney 1-Forms):** Canonical conforming finite element solver on structured triangular meshes [3], associating exactly

one tangential circuital degree of freedom with each interior triangle edge ($N_{\text{DoF}} = N_{\text{edges,int}}$).

2. **Balanced Master-Slave Hybrid (FEM-VNMM):** The proposed hybrid partitioning of Ω into a master finite-element subregion $\Omega_{\text{fem}} = [0, \pi/2] \times [0, \pi]$ and a slave meshless subregion $\Omega_{\text{vnmm}} = [\pi/2, \pi] \times [0, \pi]$, coupled along the internal interface Γ_I via non-conforming directional projection. The nodal spacing in Ω_{vnmm} is selected to balance edge and nodal densities across the interface.
3. **Pure VNMM 2D ($\mathcal{P}^1$ Complete Basis):** Full meshless point-collocation solver over Ω using the 6-node linear polynomial basis with director regularization parameter $s_{\text{div}} = 6.0$. Each active internal node supports exactly one scalar component degree of freedom ($N_{\text{DoF}} = (N_v - 2)^2$).

6.5.1. Resolution Metric and Degree-of-Freedom Equivalence

In Sections 6.2 and 6.4, the notation $N_x \times N_y$ denoted the underlying geometric Cartesian grid of points. However, when comparing edge-based finite elements against node-based meshless formulations, a direct identification of geometric vertex count introduces an artificial discrepancy.

In a two-dimensional structured triangular mesh generated by bisecting rectangular cells, the number of edges E scales with the number of vertices V according to Euler's planar graph formula [59]:

$$E \approx 3V. \tag{65}$$

Consequently, an Edge FEM mesh of $N \times N$ vertices generates approximately $N_{\text{DoF}} \approx 3(N-1)^2$ active degrees of freedom. Conversely, a node-based meshless method generates exactly one degree of freedom per internal node, $N_{\text{DoF}} = (N-2)^2 \approx V$. Comparing both methods using identical vertex grids would compare an Edge FEM problem with three times as many degrees of freedom as the meshless solver, skewing both the linear system size and the spatial resolution.

To establish a physically and computationally fair benchmark, we define an edge-equivalent nodal discretization parameter N_v for the meshless domain such that the VNMM nodal density $\rho_{\text{node}} = N_{\text{nodes}}/|\Omega|$ matches the active Edge FEM DoF density $\rho_{\text{edge}} = N_{\text{edges}}/|\Omega|$:

$$N_v = \text{round}\big(\sqrt{3}\,(N-1)\big) + 1. \tag{66}$$

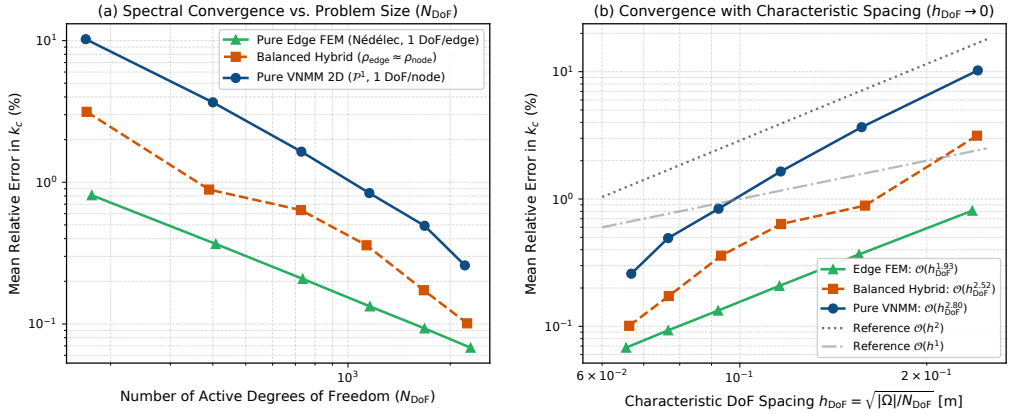


Figure 8: Comparative convergence study across 6 refinement levels ($N \in \{9, 13, 17, 21, 25, 29\}$): (a) Mean relative spectral error in cutoff wavenumber k_c (%) as a function of total active degrees of freedom N_{DoF}; (b) Asymptotic convergence rate as a function of characteristic DoF spacing $h_{\text{DoF}} = \sqrt{|\Omega|/N_{\text{DoF}}}$. Pure Edge FEM converges at $\mathcal{O}(h_{\text{DoF}}^{1.93})$, while Balanced Hybrid and Pure VNMM exhibit asymptotic convergence rates of $\mathcal{O}(h_{\text{DoF}}^{2.52})$ and $\mathcal{O}(h_{\text{DoF}}^{2.80})$, respectively.

Furthermore, to eliminate geometric grid-topology bias across all three methods, convergence is tracked as a function of the characteristic degree-of-freedom spacing:

$$h_{\text{DoF}} = \sqrt{\frac{|\Omega|}{N_{\text{DoF}}}}. \tag{67}$$

6.5.2. Convergence and Efficiency Analysis

Table 7 and Fig. 8 report the convergence history across 6 successive mesh levels ($N \in \{9, 13, 17, 21, 25, 29\}$).

Inspection of the benchmark results reveals several noteworthy numerical properties:

- **Strict Monotonicity:** With the elimination of the artificial determinant tolerance floor ($\text{Tol}_{\text{floor}}$, Eq. (44)), the local collocation determinant scales naturally as $\det(\mathbf{A}) \sim \mathcal{O}(h^4)$, allowing nearest-neighbor stencils to be preserved without unphysical support dilation. Consequently, all three formulations converge strictly monotonically.

- **Theoretical Error Ordering:** Throughout all 6 refinement levels,

Table 7: Comparative convergence of resonant wavenumber k_c for TE_z modes in a 2D PEC cavity over 6 mesh refinement levels.

Mesh Level	**Edge FEM (Nédélec 1st-kind)**			**Hybrid FEM-VNMM (Master-Slave)**			**Pure VNMM 2D ($\mathcal{P}^1$ Complete Base)**		
N –	N_{DoF} –	h_{DoF} [m]	Error (%)	N_{DoF} –	h_{DoF} [m]	Error (%)	N_{DoF} –	h_{DoF} [m]	Error (%)
9	176	0.2368	0.810	170	0.2409	3.144	169	0.2417	10.233
13	408	0.1555	0.367	390	0.1591	0.888	400	0.1571	3.667
17	736	0.1158	0.208	727	0.1165	0.636	729	0.1164	1.645
21	1160	0.0922	0.133	1134	0.0933	0.359	1156	0.0924	0.839
25	1680	0.0766	0.093	1672	0.0768	0.173	1681	0.0766	0.493
29	2296	0.0656	0.068	2243	0.0663	0.101	2209	0.0668	0.259

the accuracy satisfies the hierarchy:

$$\text{Error}_{\text{FEM}} < \text{Error}_{\text{Hybrid}} < \text{Error}_{\text{VNMM}}. \tag{68}$$

- **Asymptotic Convergence Rate and Pre-asymptotic Spectral Stiffening:** While Edge FEM exhibits the canonical second-order rate ($\mathcal{O}(h_{\text{DoF}}^{1.93})$), the Balanced Hybrid and Pure VNMM solvers exhibit steeper asymptotic slopes of 2.52 and 2.80, respectively. This enhanced descent rate reflects the higher polynomial completeness of the local $\mathcal{P}^1$ vector space, coupled with the selective spectral stiffening exerted by the Ritz-Galerkin div-curl penalty on coarser grids, which dissipates as $h \to 0$ and enables the meshless and hybrid formulations to rapidly close the accuracy gap with FEM.

- **Practical Advantages of the Hybrid Formulation:** While pure Edge FEM delivers slightly lower errors for an identical DoF count on simple regular geometries (0.068% vs 0.101% on the finest mesh), the Balanced Master-Slave Hybrid formulation offers distinct practical advantages for complex engineering problems. By confining the finite element triangulation strictly to boundary-conforming layers where tangential PEC conditions require high circulatory fidelity, the hybrid approach allows the interior domain Ω_{vnmm} to be discretized via unconstrained meshless point clouds. This substantially mitigates the computational bottleneck and quality degradation of volumetric mesh generation in moving-part, multi-scale, or topologically evolving electromagnetic devices without sacrificing interface flux continuity.

It is important to emphasize the methodological synergy between the support selection strategies: while the absolute tolerance floor ($\text{Tol}_{\text{floor}} = 1.5 \times 10^{-5}$) is indispensable to protect stencils against pathological degeneracies in stochastically perturbed or unconstrained point clouds (Section 6.3), smooth asymptotic refinement on regular grids benefits from allowing the determinant to scale naturally as $\mathcal{O}(h^4)$ to maintain compact nearest-neighbor supports. Combined with the div-curl penalty regularization ($s_{\text{div}} = 6.0$) established in Section 6.4, the global stiffness matrix $\mathbf{K}$ remains strictly well-conditioned and free of spurious spectral modes across both structured and unstructured discretization regimes.

6.6. Limitations and Scope of Applicability

While the numerical experiments in Sections 6.1–6.5 demonstrate the convergence, conditioning stability, and spectral purity of the proposed VNMM 2D formulation, a mature computational framework requires an explicit delineation of its operational scope, potential failure modes, and methodological limitations [11]. We identify six primary boundaries of applicability:

1. **Background Quadrature Integration Dependency:** Although the local vector shape function construction is completely free of topological mesh connectivity, evaluating the global Ritz-Galerkin weak form (Section 4) relies on background quadrilateral integration cells with Gauss-Legendre quadrature (EFG-style [14, 39]). Because the rational/non-polynomial meshless basis functions are non-conforming with cell boundaries, high-order quadrature (2×2 Gauss points per background cell in this work) is necessary to keep integration errors below interpolation errors [39]. For domains with highly complex, moving, or multi-scale boundaries, generating conforming background quadrature cells reintroduces a geometric pre-processing burden. Developing truly cell-free integration schemes—such as stabilized conforming nodal integration (SCNI) or meshless local Petrov-Galerkin (MLPG) subdomains [13]—remains an essential objective for fully unconstrained meshless vector solvers.
2. **Pathological Stencils, Severe Anisotropy, and Support Starvation:** The automated support qualification algorithm (Section 3.3) assumes that the candidate search neighborhood contains adequate angular and spatial diversity. Local collocation matrix degeneracies in $\mathbf{A}_{6\times6}$ manifest primarily through three geometric failure modes: (i)

near-parallel directors (≥ 4 directors with mutual angles $< 5^\circ$), which eliminate linear independence among directional projection rows; (ii) near-collinear node distributions (≥ 5 nodes aligned along a line, or extreme boundary-layer aspect ratios $\Delta x/\Delta y \gg 10$), which collapse the 2D polynomial Vandermonde submatrix; and (iii) boundary and convex corner starvation, where domain truncation restricts the accessible angular half-space. Although the adaptive k-d tree candidate search pool ($K \in [8, 16]$) successfully recovers regular stencils in unconstrained clouds, severe directional clustering necessitates wider neighborhood sampling, resulting in localized support dilation and elevated condition numbers $\kappa_2(\mathbf{A})$.

3. **Scale Dependence of Determinant Thresholding ($\mathrm{Tol}_{\mathrm{floor}}$):** As proven in Section 3.3.1, the collocation placement determinant scales with the fourth power of spatial dimension, $[\det(\mathbf{A}_{6\times6})] = [L]^4$ (Eq. (41)). Consequently, the absolute tolerance floor $\mathrm{Tol}_{\mathrm{floor}} = 1.5 \times 10^{-5}$ established in this work is calibrated for computational domains with characteristic dimensions on the order of unity ($L_{\mathrm{domain}} \sim \pi$). When transitioning to micro-scale systems (e.g., nanophotonics and microelectromechanical devices with $L \sim 10^{-6}$ m) or macro-scale engineering structures ($L \sim 10^2$ m), this threshold must be scaled dimensionally as $\mathrm{Tol}_{\mathrm{floor}} \propto L^4$, or replaced by the dimensionless scale-invariant ratio $\tau_{\mathrm{rel}} = |\det(\mathbf{A})|/h_{\mathrm{local}}^4$ to avoid false-positive stencil rejection or ill-conditioned support acceptance.
4. **Div-Curl Penalty Sensitivity and Numerical Over-Stiffening:** The Ritz-Galerkin div-curl regularization shifts non-physical gradient nullspace modes to the high-frequency spectrum for penalty values $s_{\mathrm{div}} \in [4.0, 10.0]$ without perturbing physical resonant frequencies. However, because discrete divergence at Gauss quadrature points is only weakly enforced ($\nabla \cdot \vec{E}^h \sim \mathcal{O}(h)$), setting an excessively large penalty parameter ($s_{\mathrm{div}} \gg 50$) exerts an artificial volumetric over-stiffening (locking) effect on solenoidal fields, deteriorating field convergence rates and inflating global matrix condition numbers. Conversely, taking $s_{\mathrm{div}} \to 0$ allows spurious gradient modes to re-enter the physical operational bandwidth.
5. **High-Frequency Electromagnetics and Numerical Dispersion:** The eigenvalue benchmarks presented in Section 6.4 validate the spectral behavior across the lower cavity resonant modes ($\lambda \leq 20$). In high-frequency, short-wavelength regimes ($kh \gg 1$), all discrete wave

solvers suffer from numerical dispersion and phase lag (the pollution effect) [1, 3]. For the VNMM, maintaining phase accuracy requires satisfying the standard spatial sampling criterion ($h \leq \lambda_{\mathrm{EM}}/10$). Furthermore, for electrically large domains, the penalty parameter s_{div} may require frequency-dependent scaling to ensure consistent spectral separation between physical and non-physical modes.

6. **Dimensional Scope (2D Formulation) and Non-Linear Media:** The mathematical formulation and numerical experiments in this work are developed for two-dimensional electromagnetic vector fields (xy-plane, TE_z polarization). Extending the complete vector basis to three dimensions requires a 12-dimensional polynomial space $\mathcal{P}_1 \times \mathcal{P}_1 \times \mathcal{P}_1$ ($M_{\mathrm{supp}} = 12$ support nodes per stencil) [32], which substantially expands the combinatorial candidate pool $\binom{K}{12}$ and introduces 3D surface-integral boundary representations. Additionally, extending the regularized VNMM and hybrid FEM-VNMM frameworks to non-linear eddy-current problems with ferromagnetic saturation (B–H curves) necessitates iterative Newton-Raphson linearizations and time-stepping schemes [43, 20].

7. Conclusion and Future Works

This paper has presented a comprehensive mathematical, algebraic, and variational framework for the Vector Nodal Meshless Method (VNMM 2D) applied to electromagnetic vector field problems and resonant cavity analysis. By addressing key mathematical limitations of early solenoidal vector meshless formulations, this work provides a stabilized and accurate computational framework for vector approximation on arbitrary point clouds without mesh connectivity.

7.1. Summary of Core Contributions and Key Findings

A pivotal advancement established in this work lies in overcoming the topological constraints of earlier VNMM literature [26, 31]. In foundational studies by Ortiz [26] and Gonçalves [31], support node selection relied strictly on background geometric structures, such as planar subdivisions or Delaunay triangulations, to prevent algebraic singularities and enforce interface field continuity. While effective in structured scenarios, this requirement restricted the method's flexibility and preserved a partial dependence on mesh generation tools.

The present work addresses this topological bottleneck, demonstrating that the VNMM can operate robustly on **unstructured node distributions and arbitrary vector director orientations** without requiring auxiliary planar subdivisions or background mesh connectivity for approximation.

The primary theoretical and numerical achievements of this research are summarized as follows:

1. **Transition to Mesh-Connectivity-Free Support Qualification:** By replacing pre-defined spatial subdivisions with a 4-phase dual qualification algorithm—combining k-d tree spatial queries with scale-dependent determinant thresholding—the proposed solver automatically identifies well-conditioned support sextets across unstructured node clouds. This removes background mesh connectivity requirements and enables robust operation under stochastic spatial jitter and arbitrary vector orientations.
2. **Elimination of Modal Aliasing via Complete $\mathcal{P}^1$ Vector Basis:** Transitioning from the incomplete 3-term solenoidal space $\mathcal{L}^1$ to the complete linear polynomial vector space $\mathcal{P}^1 = \mathcal{P}_1 \times \mathcal{P}_1$ ($M_{\text{supp}} = 6$ nodes) resolves all four independent spatial components of the local Jacobian tensor $\{\partial E_x/\partial x,\ \partial E_x/\partial y,\ \partial E_y/\partial x,\ \partial E_y/\partial y\}$. This removes the derivative modal aliasing mechanism, breaking the $\mathcal{O}(1)$ curl stagnation plateau of $\mathcal{L}^1$ and restoring first-order convergence $\mathcal{O}(h^{0.97})$ for the curl operator $(\nabla \times \vec{E}^h)_z$, alongside an observed convergence rate of $\mathcal{O}(h^{2.17})$ for the electric field $\vec{E}^h$ (19.8× precision gain) and a 25.9× error reduction for the field curl on the finest grid.
3. **Algebraic Stabilization via Absolute Determinant Floor:** Evaluating the scale-dependent placement determinant $|\det(\mathbf{A}_{6\times 6})|$ serves as a single, unified algebraic filter against all placement matrix failure modes. Enforcing an absolute tolerance floor ($\text{Tol}_{\text{floor}} = 1.5 \times 10^{-5}$) in the candidate search provides robust matrix conditioning ($\kappa_2(\mathbf{A}) \leq 73$, reaching ≈ 64 for the optimal threshold), maintaining monotonic error reduction across the tested refinement sequence under 25% stochastic spatial displacement and random director orientations.
4. **Spectral Purification via Div-Curl Regularization:** Because point-collocated meshless shape functions do not form an exact discrete de Rham complex at background quadrature points, unregularized formulations suffer from discrete gradient nullspace leakage ($\nabla \times (\nabla\phi) \neq \mathbf{0}$)

[3, 5]. Incorporating a Ritz-Galerkin div-curl penalty term ($s_{\text{div}} = 6.0$) shifts non-physical gradient modes to high frequencies ($\lambda > 50$), achieving spurious-free spectral pairing across the investigated TE_z operational bandwidth and restricting the mean cutoff wavenumber error to 1.00%.

5. **Conforming Master-Slave Hybrid Coupling Without Saddle-Point Indefiniteness:** By placing VNMM interface nodes at the midpoints of adjacent Whitney edge finite elements and establishing the exact dimensional scale transformation $c_k = e_k/l_k$, the conforming master-slave hybrid solver eliminates interface unknowns without auxiliary Lagrange multipliers. This avoids zero-mass saddle-point indefiniteness, preserves symmetric positive-definite (SPD) system properties under regularized boundary constraints, and prevents spurious infinite eigenvalues ($\lambda = \infty$), delivering high resonance accuracy (0.101% error on the finest mesh) with a lightweight DoF footprint.

7.2. Promising Directions for Future Research

Building upon the established 2D vector meshless foundation, several promising avenues for future research are identified:

1. **Extension to 3D Vector Nodal Meshless Formulations (3D VNMM):** Extending the complete polynomial space concept to three-dimensional vector domains ($\mathcal{P}^1$ in 3D, requiring $M_{\text{supp}} = 12$ support nodes) to solve complex 3D high-frequency waveguiding and radiation problems [32].
2. **Non-Linear Electromagnetic Applications and Motion Modeling:** Extending the regularized VNMM 2D and hybrid FEM-VNMM solvers to transient non-linear eddy-current problems, incorporating B–H iron core saturation, field-circuit coupling, and electrical machine rotor movement modeling without remeshing [43, 20].
3. **Adaptive Meshless Refinement (h- and p-Adaptivity):** Developing a posteriori local residual error estimators based on the placement determinant $|\det(\mathbf{A})|$ and local curl jump indicators to drive automatic node insertion (h-adaptivity) and local polynomial order elevation (p-adaptivity) around re-entrant corners and material interfaces.
4. **High-Performance Parallel Computing and GPU Acceleration:** Accelerating the point-based Element-Free Galerkin (EFG) Gauss integration loop and local placement matrix inversions through massively

parallel CUDA/GPU implementations, exploiting the decoupled nature of local k-d tree candidate queries.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work, the authors used Google Antigravity / Gemini in order to assist with English language editing, LaTeX code formatting, script automation for numerical data consistency, and cross-referencing verification. After using this tool, the authors carefully reviewed, verified, and edited all resulting content and take full responsibility for the integrity and accuracy of the publication.

Acknowledgements

The authors would like to thank the Brazilian research funding agencies CAPES, CNPq, and FAPEMIG for their financial support during the development of this research project.

References

[1] J.-M. Jin, The Finite Element Method in Electromagnetics, 3rd Edition, John Wiley & Sons, Hoboken, NJ, 2014.

[2] R. F. Harrington, Time-Harmonic Electromagnetic Fields, IEEE Press, Piscataway, NJ, 2001.

[3] P. Monk, Finite Element Methods for Maxwell's Equations, Oxford University Press, Oxford, 2003. doi:10.1093/acprof:oso/9780198508885.001.0001.

[4] K. Yee, Numerical solution of initial boundary value problems involving Maxwell's equations in isotropic media, IEEE Transactions on Antennas and Propagation 14 (3) (1966) 302–307. doi:10.1109/TAP.1966.1138693.

[5] A. Bossavit, Solving Maxwell equations in a closed cavity, and the question of 'spurious modes', IEEE Transactions on Magnetics 26 (2) (1990) 702–705. doi:10.1109/20.106414.

[6] D. Boffi, P. Fernandes, L. Gastaldi, I. Perugia, Computational models of electromagnetic resonators: Analysis of edge element approximation, SIAM Journal on Numerical Analysis 36 (4) (1999) 1264–1290. doi:10.1137/S003614299731853X.

[7] J.-C. Nédélec, Mixed finite elements in $\mathbb{R}^3$, Numerische Mathematik 35 (3) (1980) 315–341. doi:10.1007/BF01396415.

[8] A. Bossavit, Computational Electromagnetism: Variational Formulations, Complementarity, Edge Elements, Academic Press, San Diego, CA, 1998.

[9] S. R. Idelsohn, E. Oñate, To mesh or not to mesh... that is the question..., Computer Methods in Applied Mechanics and Engineering 195 (37–40) (2006) 4681–4696. doi:10.1016/j.cma.2005.11.006.

[10] G.-R. Liu, Mesh Free Methods: Moving Beyond the Finite Element Method, 2nd Edition, CRC Press, Boca Raton, FL, 2009.

[11] J. Slak, G. Kosec, Refined meshless local strong form solution of cauchy-navier equation on an irregular domain, Engineering Analysis with Boundary Elements 90 (2018) 1–11. doi:10.1016/j.enganabound.2018.01.001.

[12] G.-R. Liu, Y.-T. Gu, An Introduction to Meshfree Methods and Their Programming, Springer, Dordrecht, Netherlands, 2005. doi:10.1007/1-4020-3228-5.

[13] S. N. Atluri, S.-P. Shen, The Meshless Local Petrov-Galerkin (MLPG) Method, Tech Science Press, Encino, CA, 2002.

[14] T. Belytschko, Y. Y. Lu, L. Gu, Element-free Galerkin methods, International Journal for Numerical Methods in Engineering 37 (2) (1994) 229–256. doi:10.1002/nme.1620370205.

[15] S. N. Atluri, T. Zhu, A new meshless local Petrov-Galerkin (MLPG) approach in computational mechanics, Computational Mechanics 22 (2) (1998) 117–127. doi:10.1007/s004660050346.

[16] G.-R. Liu, Y.-T. Gu, A point interpolation method for two-dimensional solids, International Journal for Numerical Methods in Engineering

50 (4) (2001) 937–951. doi:10.1002/1097-0207(20010210)50:4<937::AID-NME62>3.0.CO;2-X.

[17] J. J. Monaghan, Smoothed particle hydrodynamics, Annual Review of Astronomy and Astrophysics 30 (1) (1992) 543–574. doi:10.1146/annurev.aa.30.090192.002551.

[18] S. A. Viana, R. C. Mesquita, Moving least square reproducing kernel method for electromagnetic field computation, IEEE Transactions on Magnetics 35 (3) (1999) 1372–1375. doi:10.1109/20.767218.

[19] G. F. Parreira, E. J. Silva, A. R. Fonseca, R. C. Mesquita, The Element-Free Galerkin method in three-dimensional electromagnetic problems, IEEE Transactions on Magnetics 42 (4) (2006) 711–714. doi:10.1109/TMAG.2006.872014.

[20] E. H. R. Coppoli, R. C. Mesquita, R. S. Silva, Induction machines modeling with meshless methods, IEEE Transactions on Magnetics 48 (2) (2012) 847–850. doi:10.1109/TMAG.2011.2174211.

[21] V. Cingoski, N. Miyamoto, H. Yamashita, Element-free Galerkin method for electromagnetic field computations, IEEE Transactions on Magnetics 34 (5) (1998) 3236–3239. doi:10.1109/20.717759.

[22] A. R. Fonseca, S. A. Viana, E. J. Silva, R. C. Mesquita, Imposing boundary conditions in the meshless local Petrov–Galerkin method, IET Science, Measurement & Technology 2 (6) (2008) 387–394. doi:10.1049/iet-smt:20080082.

[23] W. L. Nicomedes, R. C. Mesquita, F. J. S. Moreira, The meshless local Petrov–Galerkin method in two-dimensional electromagnetic wave analysis, IEEE Transactions on Antennas and Propagation 60 (4) (2012) 1957–1968. doi:10.1109/TAP.2012.2186223.

[24] W. L. Nicomedes, Mixed meshfree methods in computational electromagnetism: Mathematical foundations and problems in wave scattering, Ph.D. thesis, Programa de Pós-Graduação em Engenharia Elétrica, Universidade Federal de Minas Gerais (UFMG), Belo Horizonte, Brasil (2015).

[25] N. Z. Lima, Métodos sem malha aplicados ao eletromagnetismo: Formas fracas enfraquecidas e funções de forma vetoriais, Ph.D. thesis, Programa de Pós-Graduação em Engenharia Elétrica, Universidade Federal de Minas Gerais (UFMG), Belo Horizonte, Brasil (2016).

[26] L. A. G. Ortiz, Método sem malha nodal utilizando funções de forma vetoriais baseadas em H(curl), Ph.D. thesis, Programa de Pós-Graduação em Engenharia Elétrica, Universidade Federal de Minas Gerais (UFMG), Belo Horizonte, Brasil (2023).

[27] N. Z. Lima, R. C. Mesquita, Edge meshless method applied to vector electromagnetic problems, IEEE Transactions on Magnetics 53 (6) (2017) 1–4. doi:10.1109/TMAG.2017.2653769.

[28] P. M. I. Silva, R. C. Mesquita, N. Z. Lima, New vectorial $H(\text{curl})$ shape functions for the edge meshless method, IET Science, Measurement & Technology 12 (6) (2018) 821–828. doi:10.1049/iet-smt.2018.0111.

[29] L. A. G. Ortiz, R. C. Mesquita, N. Z. Lima, Creating higher order vector shape functions based on $H(\text{curl})$ for the edge meshless method, Journal of Electromagnetic Waves and Applications 35 (11) (2021) 1511–1528. doi:10.1080/09205071.2021.1901787.

[30] L. A. G. Ortiz, R. C. Mesquita, N. Z. Lima, Nodal meshless method with vectorial shape functions based on H(curl), IEEE Transactions on Magnetics 57 (6) (2021) 1–4. doi:10.1109/TMAG.2021.3061526.

[31] B. M. F. Gonçalves, Construção das funções de forma 3-D para o método sem malha nodal vetorial, Ph.D. thesis, Programa de Pós-Graduação em Engenharia Elétrica, Universidade Federal de Minas Gerais (UFMG), Belo Horizonte, Brasil (2025).

[32] B. M. F. Gonçalves, E. J. Silva, R. C. Mesquita, Vector nodal meshless method for 3-D applications, IEEE Transactions on Magnetics 59 (5) (2023) 1–4. doi:10.1109/TMAG.2022.3233527.

[33] B. M. F. Gonçalves, E. J. Silva, R. C. Mesquita, M. M. Afonso, B. Ramdane, Periodic boundary conditions in Vector Nodal Meshless Method, IEEE Transactions on Magnetics 62 (2026). doi:10.1109/TMAG.2025.3609622.

[34] S. Li, W. K. Liu, Meshfree Particle Methods, Springer Science & Business Media, Berlin, Germany, 2007. doi:10.1007/978-3-540-71471-2.

[35] J.-C. Nédélec, A new family of mixed finite elements in $\mathbb{R}^3$, Numerische Mathematik 50 (1) (1986) 57–81. doi:10.1007/BF01389668.

[36] J. L. Bentley, Multidimensional binary search trees used for associative searching, Communications of the ACM 18 (9) (1975) 509–517. doi:10.1145/361002.361007.

[37] G. F. Parreira, A. R. Fonseca, A. C. Lisboa, E. J. Silva, R. C. Mesquita, Efficient algorithms and data structures for Element-Free Galerkin method, IEEE Transactions on Magnetics 42 (4) (2006) 659–662. doi:10.1109/TMAG.2006.871432.

[38] G. E. Fasshauer, Meshfree Approximation Methods with MATLAB, Vol. 6 of Interdisciplinary Mathematical Sciences, World Scientific Publishing, Singapore, 2007. doi:10.1142/6437.

[39] J. Dolbow, T. Belytschko, Numerical integration of Galerkin weak forms in meshfree methods, International Journal for Numerical Methods in Engineering 46 (5) (1999) 657–683. doi:10.1007/s004660050403.

[40] Y. Saad, Numerical Methods for Large Eigenvalue Problems, revised Edition, SIAM, Philadelphia, PA, 2011. doi:10.1137/1.9781611970739.

[41] R. B. Lehoucq, D. C. Sorensen, C. Yang, ARPACK Users' Guide: Solution of Large-Scale Eigenvalue Problems with Implicitly Restarted Arnoldi Methods, SIAM, Philadelphia, PA, 1998. doi:10.1137/1.9780898719628.

[42] P. Virtanen, R. Gommers, T. E. Oliphant, et al., SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python, Nature Methods 17 (3) (2020) 261–272. doi:10.1038/s41592-019-0686-2.

[43] I. A. Tsukerman, Overlapping finite elements for problems with movement, IEEE Transactions on Magnetics 28 (5) (1992) 2247–2249. doi:10.1109/20.179458.

[44] R. El Kadmiri, Y. Belaasilia, A. Timesli, M. S. Kadiri, A hybrid algorithm using the FEM-MESHLESS method to solve nonlinear structural

problems, Engineering Analysis with Boundary Elements 140 (2022) 531–543. doi:10.1016/j.enganabound.2022.04.018.

[45] A. Quarteroni, A. Valli, Domain Decomposition Methods for Partial Differential Equations, Oxford University Press, Oxford, UK, 1999.

[46] B. Smith, P. Bjørstad, W. Gropp, Domain Decomposition: Parallel Multilevel Methods for Elliptic Partial Differential Equations, Cambridge University Press, Cambridge, UK, 1996.

[47] A. Toselli, O. Widlund, Domain Decomposition Methods — Algorithms and Theory, Vol. 34 of Springer Series in Computational Mathematics, Springer, Berlin, Germany, 2005. doi:10.1007/b137868.

[48] J. Nitsche, Über ein Variationsprinzip zur lösung von Dirichlet-Problemen bei Verwendung von Teilräumen, die die Randbedingungen nicht erfüllen, Abhandlungen aus dem Mathematischen Seminar der Universität Hamburg 36 (1) (1971) 9–15. doi:10.1007/BF02995904.

[49] S. Fernández-Méndez, A. Huerta, Imposing essential boundary conditions in mesh-free methods, Computer Methods in Applied Mechanics and Engineering 193 (12) (2004) 1257–1275. doi:https://doi.org/10.1016/j.cma.2003.12.019.

[50] F. Brezzi, On the existence, uniqueness and approximation of saddle-point problems arising from Lagrange multipliers, RAIRO. Analyse Numérique 8 (R2) (1974) 129–151.

[51] C. Bernardi, Y. Maday, A. Patera, A new non-conforming approach to domain decomposition: The mortar element method, in: H. Brézis, J. L. Lions (Eds.), Nonlinear Partial Differential Equations and Their Applications, Pitman, 1994, pp. 13–51.

[52] A. Buffa, Y. Maday, F. Rapetti, A mortar element method for $H(\mathrm{curl})$-conforming windowing problems, SIAM Journal on Numerical Analysis 39 (1) (2001) 81–101. doi:10.1137/S0036142999357968.

[53] J. M. Melenk, I. Babuška, The partition of unity finite element method: Basic theory and applications, Computer Methods in Applied Mechanics and Engineering 139 (1-4) (1996) 289–314. doi:10.1016/S0045-7825(96)01087-0.

[54] Y. Krongauz, T. Belytschko, Enforcement of essential boundary conditions in meshless approximations using finite elements, Computer Methods in Applied Mechanics and Engineering 131 (1-2) (1996) 133–145. doi:10.1016/0045-7825(95)00954-X.

[55] A. Hansbo, P. Hansbo, An unfitted finite element method, based on Nitsche's formulation, for elliptic interface problems, Computer Methods in Applied Mechanics and Engineering 191 (47-48) (2002) 5537–5552. doi:10.1016/S0045-7825(02)00524-8.

[56] F. Ben Belgacem, The mortar finite element method with Lagrange multipliers, Numerische Mathematik 84 (2) (1999) 173–197. doi:10.1007/s002110050468.

[57] I. Babuška, J. M. Melenk, The partition of unity method, International Journal for Numerical Methods in Engineering 40 (4) (1997) 727–758. doi:10.1002/(SICI)1097-0207(19970228)40:4<727::AID-NME86>3.0.CO;2-N.

[58] T. Belytschko, D. Organ, Y. Krongauz, A coupled finite element-element-free Galerkin method, Computational Mechanics 17 (3) (1995) 186–195. doi:10.1007/BF00364080.

[59] M. de Berg, O. Cheong, M. van Kreveld, M. Overmars, Computational Geometry: Algorithms and Applications, 3rd Edition, Springer-Verlag, Berlin, Heidelberg, 2008. doi:10.1007/978-3-540-77974-2.